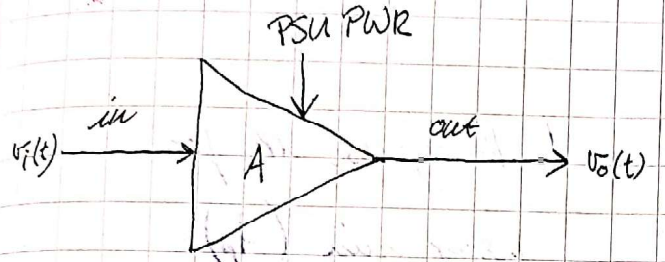
Chapter 1.4 Amplifiers

→ Amplifiers are active, transistor-based devices that amplify the voltage or current of an input signal to deliver it to a load.



A - amplification factor - gain

$$v_o(t) = A \cdot v_i(t)$$

→ Amps are needed when some sources generate weak signals (low voltage or current); ex: acoustic/electric instruments, RF antennas, phototransistors, thermocouples

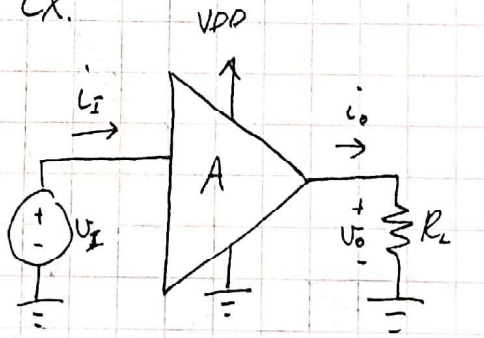
→ Ideal & amplifiers are linear ($A = const$) with respect to input voltage amplitude, input frequency, temperature.

→ Types of amplifier gain

$$A \equiv \frac{\text{output}}{\text{input}}$$

- voltage gain : $A_V \equiv \frac{v_o}{v_i}$ $\left[\frac{V}{V}\right]$
- current gain : $A_i \equiv \frac{i_o}{i_i}$ $\left[\frac{A}{A}\right]$
- power gain : $A_p \equiv \frac{P_o}{P_i}$ $\left[\frac{W}{W}\right]$

Ex.



$$\left.\begin{aligned} P_I &= U_I \cdot i_I \\ P_o &= v_o \cdot i_o \end{aligned}\right\} \Rightarrow A_p = \frac{P_o}{P_I} = \frac{U_o \cdot i_o}{U_I \cdot i_I} = A_V \cdot A_i$$

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→ Decibels: a logarithmic unit that indicates the ratio of physical quantities. They're useful when dealing with a wide range of signals as it allows both very small and very large signals to be visible on one scale.

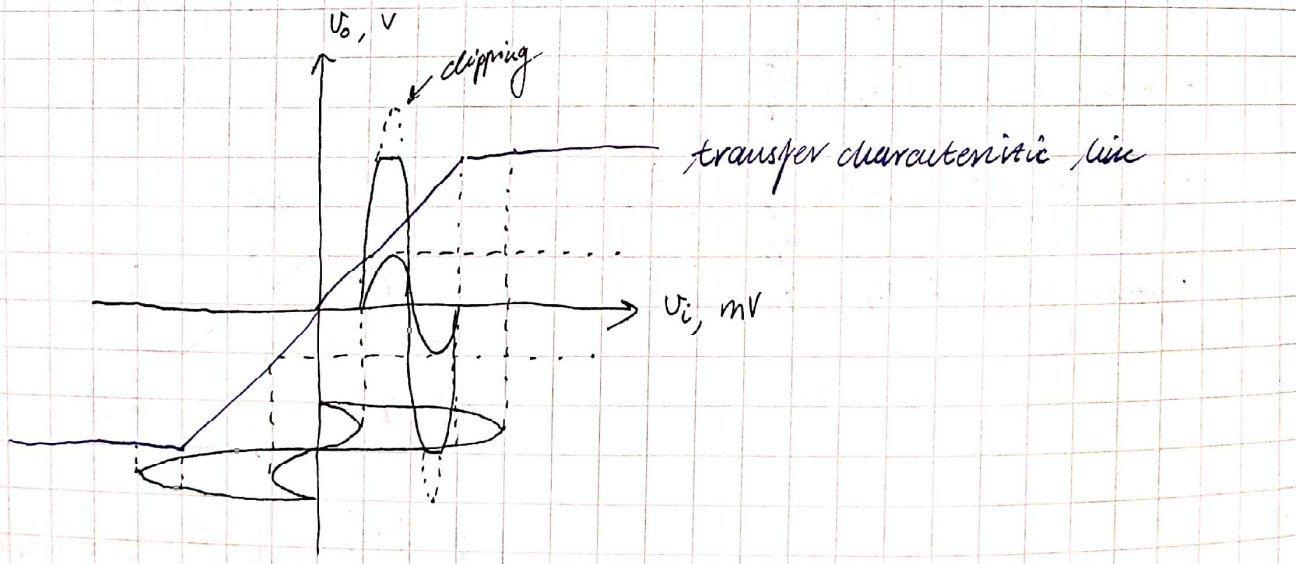
- $A_{v_{dB}} \equiv 20 \log_{10} |A_v|$ [dB]
- $A_{i_{dB}} \equiv 20 \log_{10} |A_i|$ [dB]
- $A_{P_{dB}} \equiv 10 \log_{10} |A_p|$ [dB]

! Always use dB for power gain (A_p)

→ Common gain values from dB to actual value:

- $A_{v_{dB}} = 20dB \rightarrow A_v = 10 \frac{V}{V}$
- $A_{v_{dB}} = 40dB \rightarrow A_v = 100 \frac{V}{V}$
- $A_{v_{dB}} = 3dB \rightarrow A_v = \sqrt{2} \frac{V}{V}$
- $A_{v_{dB}} = 6dB \rightarrow A_v = 2 \frac{V}{V}$
- $A_{P_{dB}} = 10dB \rightarrow A_p = 10 \frac{W}{W}$

→ Amplifier transfer characteristic graph:



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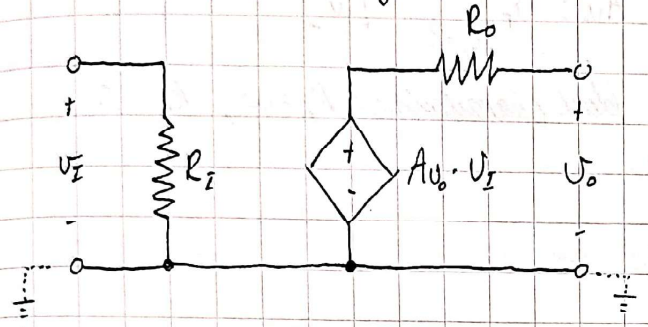
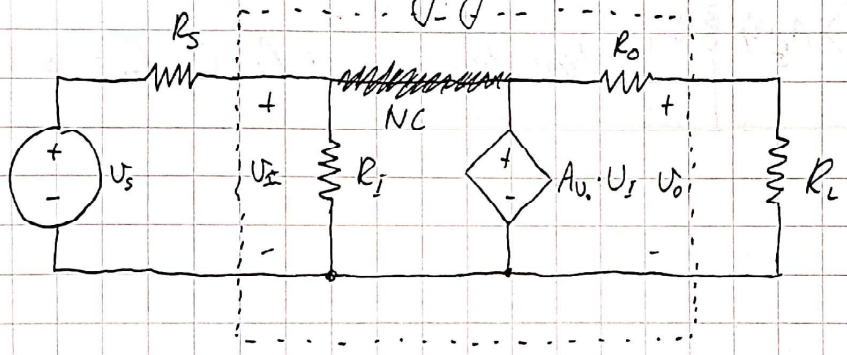
Chapter 1.5 Amplifier circuit model

→ Voltage amplifier circuit model: emulates / mimics the behavior of an amplifier using basic circuit elements



A_{v_o} : open circuit voltage gain

Ex. Find overall voltage gain $= \frac{v_o}{v_s}$



step 1) find v_o in terms of v_I:

$$v_o = A_{v_o} \cdot v_I \cdot \frac{R_L}{R_L + R_o} \Rightarrow \frac{v_o}{v_I} = A_{v_o} \cdot \frac{R_L}{R_L + R_o}$$

step 2) find overall voltage gain, ~~find v_I in terms of v_s~~:

$$v_I = v_s \cdot \frac{R_I}{R_I + R_s}$$

Using formula above: $v_o = A_{v_o} \cdot v_I \cdot \frac{R_L}{R_L + R_o}$, plug in v_I

$$\Rightarrow v_o = A_{v_o} \cdot v_s \cdot \frac{R_I}{R_I + R_s} \cdot \frac{R_L}{R_L + R_o}, \text{ divide by } v_s$$

$$\Rightarrow \frac{v_o}{v_s} = A_{v_o} \cdot \frac{R_I}{R_I + R_s} \cdot \frac{R_L}{R_L + R_o}$$

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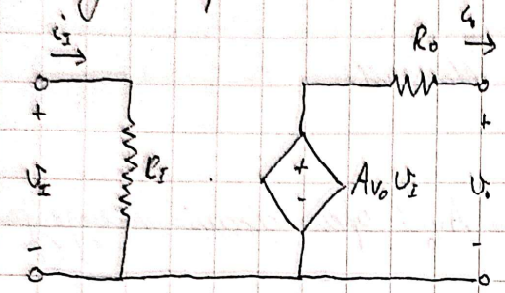
Other model types:

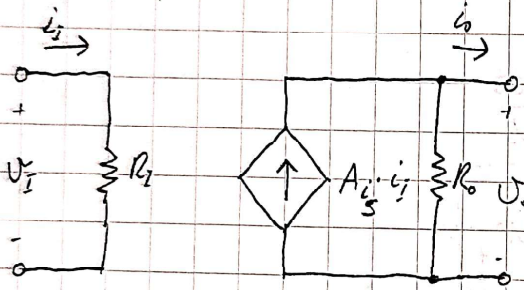
→ voltage amp:



$$A_{V_o} = \left. \frac{U_o}{U_I} \right|_{i_I = 0} \quad \left[\frac{V}{V} \right]$$

Ideal characteristic: $R_I = \infty$, $R_o = 0$

→ current amp:



$$A_{i_s} = \left. \frac{i_o}{i_I} \right|_{U_o = 0} \quad \left[\frac{A}{A} \right]$$

Ideal characteristic: $R_i = 0$, $R_o = \infty$

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Chapter 1.6 Frequency response of amplifiers

Complex numbers reminder:

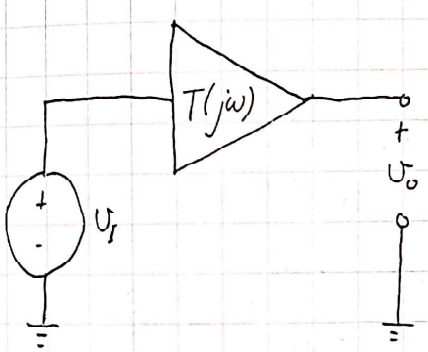
$z = a + jb$, $z = |z| \angle \theta$, $z^* = a \mp jb$ for $z = a \pm jb$

$|z| = \sqrt{a^2 + b^2}$, $\theta = \arctan\left(\frac{b}{a}\right)$, $|z| = \sqrt{z \cdot z^*}$

→ Ideal amp: $v_o(t) = A_v \cdot v_i(t)$ where A_v = const.

→ Real amp: $v_o(t) = A_v(\omega) \cdot v_i(t)$ where $A_v(\omega) \neq$ const.

→ Causes of amp non-linearity: v_i too high (causes clipping), temperature (impairs amp's performance), high frequency changes (A_v not constant at all frequencies)



$v_i = V_i \sin(\omega t) \Rightarrow v_o = V_o \sin(\omega t + \phi)$ (phase)

↑ magnitude varies with ω

$T(j\omega)$: amplifier transfer function

$T(j\omega) \equiv \frac{V_o(j\omega)}{V_i(j\omega)}$ ← both phasors

$|T(j\omega)| = \frac{V_o}{V_i}$; $\phi = \angle T(j\omega)$

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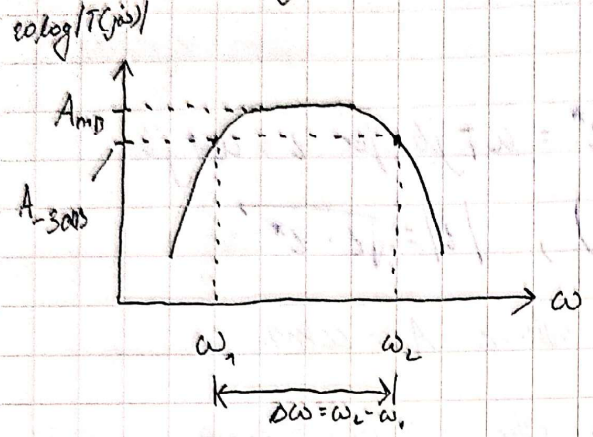
Typical frequency response curve:



A_{mB} : mid band gain

ω_1 : lower cutoff frequency

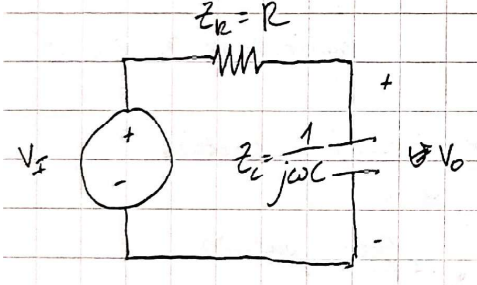
ω_2 : higher cutoff frequency

$\Delta\omega = \omega_2 - \omega_1$: bandwidth

→ Consider 3 regions:

- $\omega \ll \omega_1 \longrightarrow V_o$ becomes very small
- $\omega \gg \omega_2 \longrightarrow V_o$ becomes very small
- $\omega \in [\omega_1, \omega_2] \rightarrow |T(j\omega)| \approx A_{mB}$; constant, flat frequency response

Ex. Find $T(j\omega)$ of a simple filter:



$$V_o(j\omega) = V_i(j\omega)\frac{Z_C}{Z_C + Z_R} = V_i \cdot \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R}$$

$$T(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R} \cdot \frac{j\omega C}{j\omega C} = \frac{1}{1 + j\omega RC}$$

$\omega_0 = RC$; $\Rightarrow T(j\omega) = \frac{1}{1 + j\frac{\omega}{\omega_0}}$

$$\Rightarrow |T(j\omega)| = \sqrt{\frac{1}{1 + j\frac{\omega}{\omega_0}} \cdot \frac{1}{1 - j\frac{\omega}{\omega_0}}} = \frac{1}{\sqrt{1 + \frac{\omega^2}{\omega_0^2}}}$$

3 regions:

- $\omega \to 0 : |T(j\omega)| = \frac{1}{\sqrt{1 + \frac{0}{\omega_0^2}}} = 1$
- $\omega \to \omega_0 : |T(j\omega)| = \frac{1}{\sqrt{1 + \frac{\omega_0^2}{\omega_0^2}}} = \frac{1}{\sqrt{2}}$
- $\omega \to \infty : |T(j\omega)| = \frac{1}{\sqrt{1 + \frac{\omega^2}{\omega_0^2}}} = 0$

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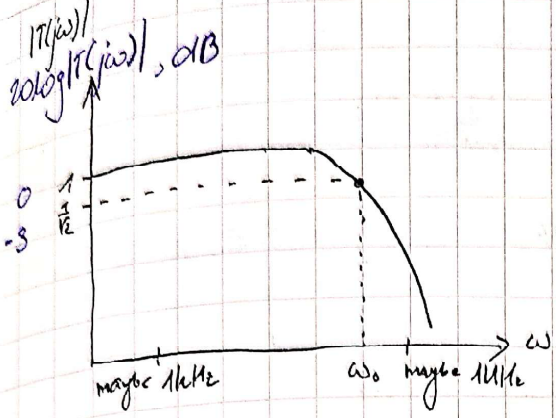
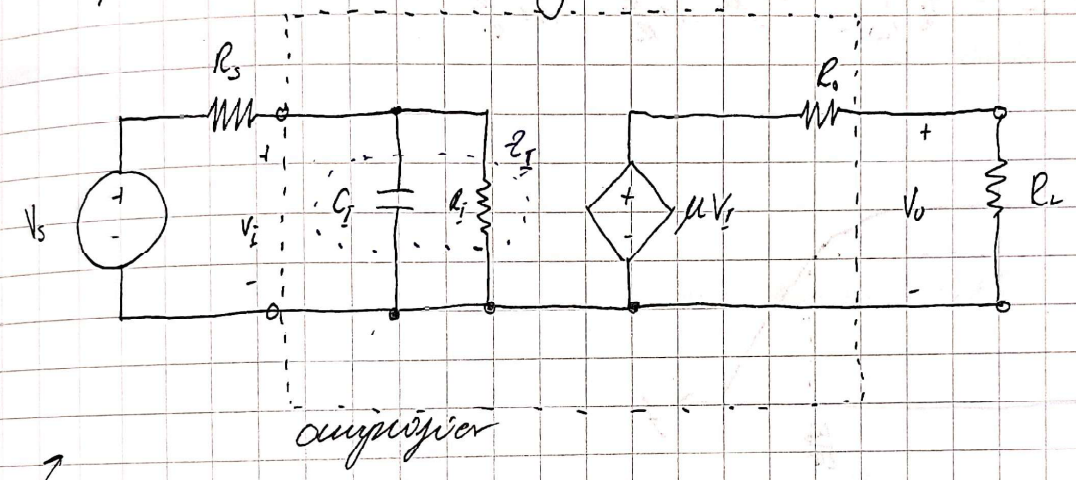


$20\log(1) = 0\,dB$

$20\log\left(\frac{1}{\sqrt{2}}\right) = -3\,dB$

A modified, more realistic voltage amplifier model (example)



Ex. Find the transfer function $T(j\omega)$ of the amplifier and plot it.

① $Z_I = R_I \,/\!/\, Z_{C_I} = R_I \,/\!/\, \frac{1}{j\omega C_I} = \frac{\frac{R_I}{j\omega C_I}}{\frac{1}{j\omega C_I} + R_I} = \frac{R_I}{1 + j\omega R_I C_I}$

② find $\frac{V_I}{V_s}$: $V_I = V_s \cdot \frac{Z_I}{Z_I + R_s} = V_s \cdot \frac{\frac{R_I}{1+j\omega R_I C_I}}{\frac{R_I}{1+j\omega R_I C_I} + R_s} \Rightarrow \frac{V_I}{V_s} = \dots = \frac{1}{1 + \frac{R_s}{R_I}} \cdot \frac{1}{1 + j\omega C_I \left(\frac{R_s R_I}{R_I + R_s}\right)}$

③ find $\frac{V_o}{V_I}$: $V_o = \mu V_I \frac{R_L}{R_L + R_o} \Rightarrow \frac{V_o}{V_I} = \mu \frac{R_L}{R_L + R_o} = \mu \frac{1}{1 + \frac{R_o}{R_L}}$

④ find $\frac{V_o}{V_s} = \frac{V_o}{V_I} \cdot \frac{V_I}{V_s} = \mu \underbrace{\frac{1}{1 + \frac{R_o}{R_L}} \cdot \frac{1}{1 + \frac{R_s}{R_I}}}_{\text{const.}} \cdot \underbrace{\frac{1}{1 + j\frac{\omega}{\omega_0}}}_{\text{LPF transfer function}}$ where $\omega_0 = \frac{1}{C_I \left(\frac{R_s R_I}{R_s + R_I}\right)}$

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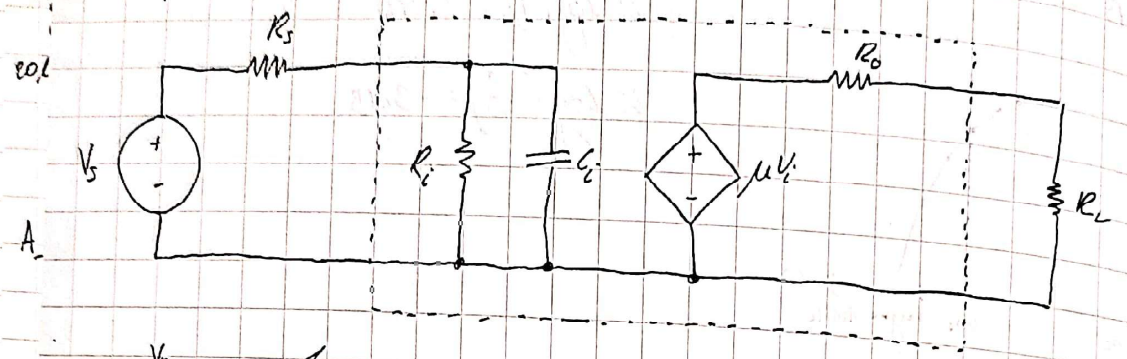
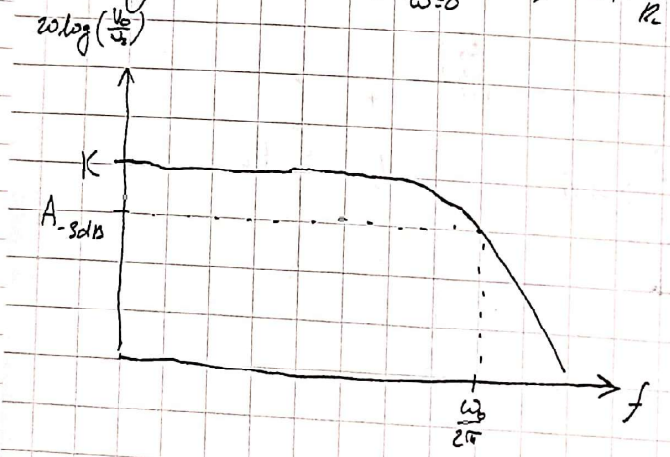
Amplifier LPF response:



$$\frac{V_o}{V_s} = \mu \cdot \frac{1}{1+\frac{R_o}{R_L}} \cdot \frac{1}{1+\frac{R_s}{R_i}} \cdot \frac{1}{1+j\frac{\omega}{\omega_0}} \Rightarrow \omega_0 = \frac{1}{C_i\left(\frac{R_s R_i}{R_s + R_i}\right)}$$

DC gain: $K \equiv \left.\frac{V_o}{V_s}\right|_{\omega=0} = \mu \cdot \frac{1}{1+\frac{R_i}{R_L}} \cdot \frac{1}{1+\frac{R_s}{R_i}}$

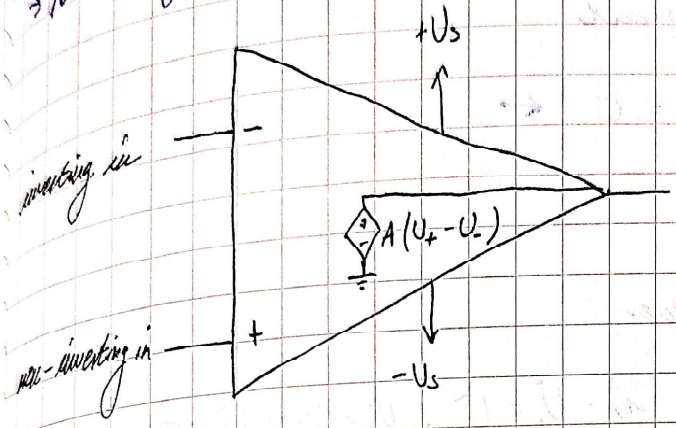


Quiz 1

Chapter 2.1: Ideal op-amp

→ An op-amp is an analog amplifier IC

→ It's a high gain differential amplifier (A between 10^5 and 10^6)



$V_o = A(V_+ - V_-)$

single-ended output

3 cases:

- $V_+ > V_- \rightarrow V_o > 0V$
- $V_+ < V_- \rightarrow V_o < 0V$

} in rail-to-rail it's V_{DD} or V_{SS}

- $V_+ = V_- \rightarrow V_o \approx 0V$ (ideally $V_o = 0V$)

CMRR (common mode ($V_+ = V_-$) rejection ratio) is an important characteristic

Ideal op-amp characteristics:

1) Infinite input impedance ($i_+ = 0$, $i_- = 0$)
2) Zero output impedance
3) Zero common mode gain
4) Infinite open loop gain (A)
5) Infinite bandwidth

Golden rules of an ideal op-amp

1) The output attempts to make the input voltage difference zero

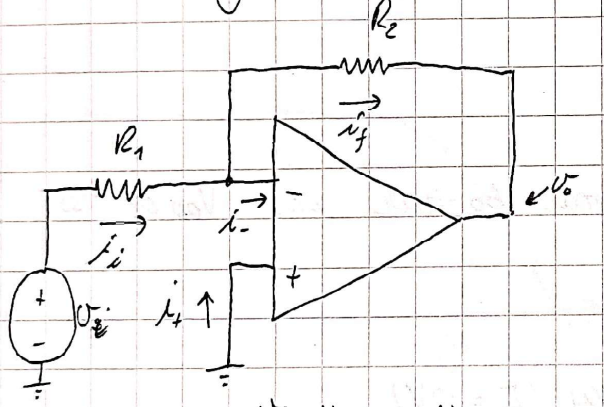
$$\rightarrow \quad v_+ = v_- \quad \leftarrow$$

2) The inputs draw zero current

$$\rightarrow \quad i_+ = i_- = 0 \quad \leftarrow$$

Chapter 2.2

The inverting op-amp amplifier



1) $v_+ = v_-$, $v_+ = 0V \Rightarrow v_- = 0V$

2) $i_+ = i_- = 0 \Rightarrow i_i = i_f$

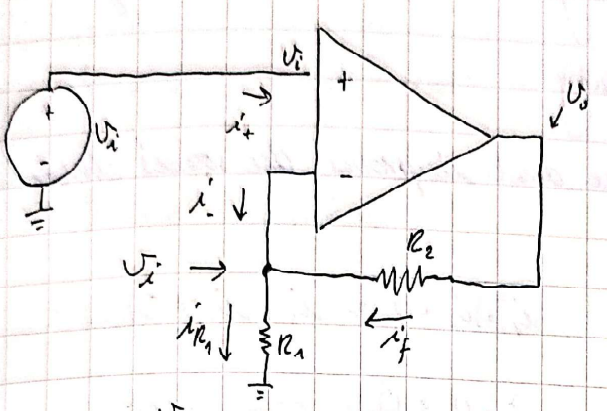
$$i_i: \quad i_i = \frac{v_i - v_-}{R_1} = \frac{v_i}{R_1}$$

$$i_f: \quad i_f = \frac{v_- - v_o}{R_2} = -\frac{v_o}{R_2}$$

$$\Rightarrow i_i = i_f \Rightarrow v_o = -\frac{R_2}{R_1} v_i$$

→ non-infinite input resistance

The non-inverting op-amp amplifier



1) $v_+ = v_s$, $v_+ = v_i \Rightarrow v_- = v_i$

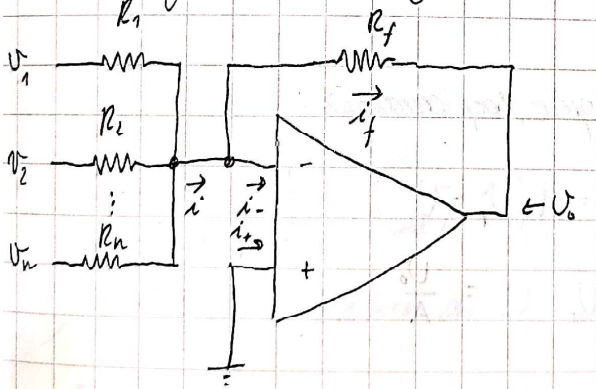
2) $i_+ = i_- = 0 \Rightarrow i_{R_1} = i_f$

$$\left.\begin{aligned} i_{R_1} &= \frac{v_x}{R_1} \\ i_f &= \frac{v_o - v_x}{R_2} \end{aligned}\right\} \; i_+ = i_- = 0 \Rightarrow i_f = i_{R_1} \Rightarrow \frac{v_x}{R_1} = \frac{v_o - v_x}{R_2} \Rightarrow v_o = v_x\left(1 + \frac{R_2}{R_1}\right)$$

→ can't attenuate

→ has infinite input impedance

The weighted inverting summer:



1) $v_+ = v_-$, $v_+ = 0 \Rightarrow v_- = 0$

2) $i_+ = i_- = 0 \Rightarrow i_f = i$

$$i = i_1 + i_2 + \dots + i_n = \frac{v_1 - 0}{R_1} + \frac{v_2 - 0}{R_2} + \dots + \frac{v_n - 0}{R_n} = \frac{v_1}{R_1} + \frac{v_2}{R_2} + \dots + \frac{v_n}{R_n}$$

$$i_f = \frac{v_- - v_o}{R_f} = -\frac{v_o}{R_f}$$

$$i = i_f \Rightarrow -\frac{v_o}{R_f} = \frac{v_1}{R_1} + \frac{v_2}{R_2} + \dots + \frac{v_n}{R_n} \Rightarrow v_o = -R_f\left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \dots + \frac{v_n}{R_n}\right)$$

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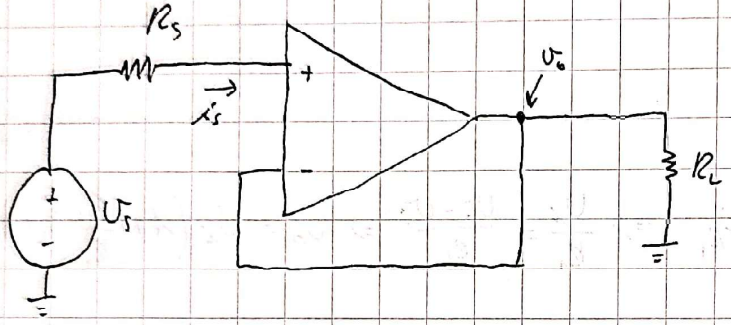
Voltage follower / buffer

→ Useful when your input source is very weak and even the slightest load drops the voltage significantly

→ Provides an infinite input impedance and improves the source's drive



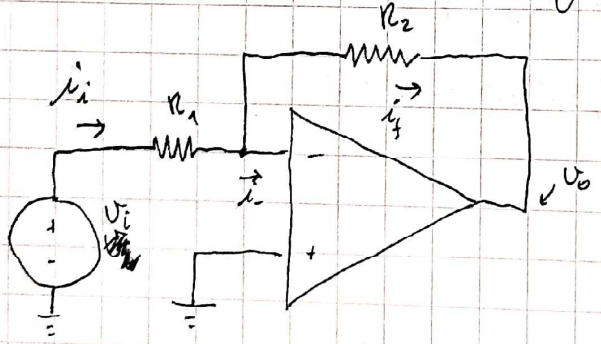
$i_+ = i_- = 0 \Rightarrow i_s = i_- = 0$

$v_+ = v_- = v_o$

$v_+ = v_s - v_{R_s} = v_s$, $v_{R_s} = 0$ since $i_s = 0$

$\Rightarrow v_o = v_+ \Rightarrow v_o = v_s$

Effect of ~~infinite~~ finite open loop gain



In open loop condition:

$v_o = A(v_+ - v_-)$

$\Rightarrow v_+ - v_- = \frac{v_o}{A}$ $\quad v_+ \neq v_-$

$i_i = \frac{v_s - v_-}{R_1}$

$v_+^{\,0} - v_- = \frac{v_o}{A}$

$\Rightarrow i_i = \frac{v_i + \frac{v_o}{A}}{R_1}$

$i_f = \frac{v_- - v_o}{R_2}$ ∧ $v_+^{\,0} - v_- = \frac{v_o}{A}$

$\Rightarrow i_f = \frac{-\frac{v_o}{A} - v_o}{R_2}$

v_+ grounded

$i_- = 0 \Rightarrow i_1 = i_f$

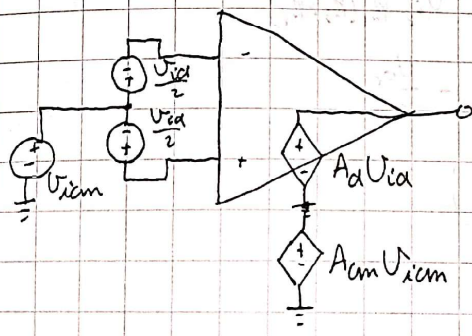
$\Rightarrow \frac{U_i + \frac{U_o}{A}}{R_1} = \frac{-\frac{U_o}{A} - U_i}{R_2} \Rightarrow \dots \Rightarrow \frac{U_o}{U_i} = -\frac{A R_2}{R_2 + R_1(1+A)}$ (closed loop gain)

$\frac{U_o}{U_i}\Big|_{A \to \infty} = -\frac{\infty R_2}{R_2 + R_1(1+\infty)} \overset{\infty}{=} -\frac{R_2}{R_1}$

Difference amplifier

- differential input: $U_{id} = U_+ - U_-$ $\quad U_o = A(U_+ - U_-)$
- common mode input: average of two inputs: $U_{icm} = \frac{U_+ + U_-}{2}$

$\Rightarrow U_- = U_{icm} - \frac{U_{id}}{2}$, $U_+ = U_{ icm} + \frac{U_{id}}{2}$


$\frac{U_{id}}{2}$
$\frac{U_{id}}{2}$
U_{icm}
$A_d U_{id}$
$A_{cm} U_{icm}$
U_o

Ideally, common mode gain should be as small as possible

CMRR: common mode rejection ratio

$CMRR = 20 \log \frac{|A_d|}{|A_{cm}|}$ [dB] typically 60dB - 120dB

Differential amplifiers are needed when a source (like a thermocouple) is floating.

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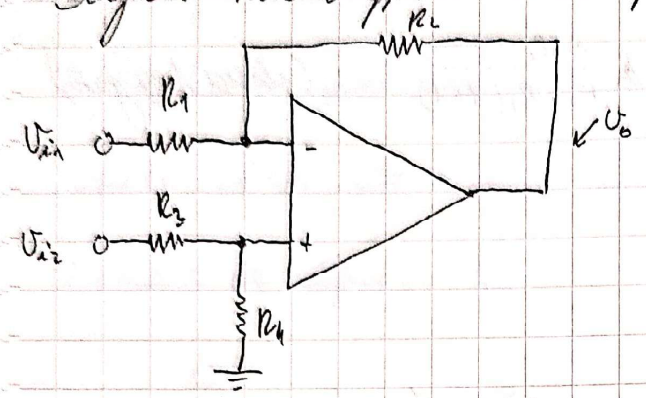
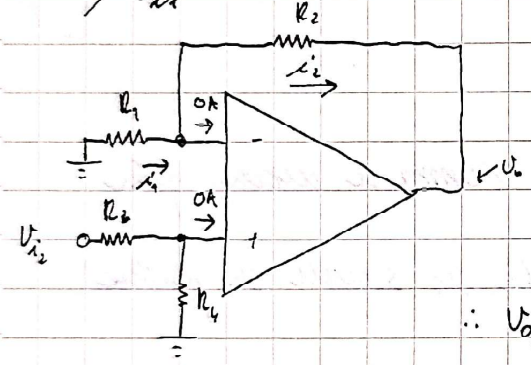
Single ended differential amplifier



Canceling out common mode gain can be done by choosing right resistor values

Using superposition:

1) $V_{i1} = 0V$



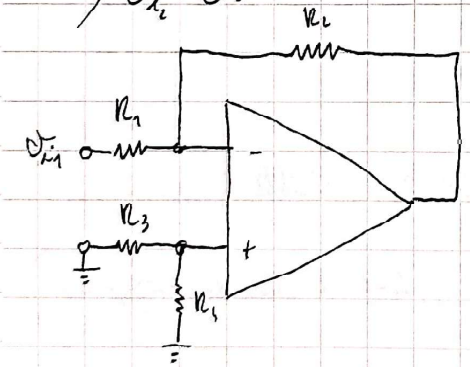
$V_- = V_+ \Rightarrow V_1 = V_{i2} \frac{R_4}{R_3+R_4} = V_-$

$i_1 = i_2$ ✓ sub for V_- $\Rightarrow i_1 = -V_{i2} \frac{R_4}{R_1(R_3+R_4)}$

$i_2 = V_{i2} \frac{R_4}{R_3(R_4+R_3)} - \frac{V_{o2}}{R_2}$

$\therefore V_{o2} = V_{i2} \frac{R_4}{R_3+R_4}\left(1+\frac{R_2}{R_1}\right)$

2) $V_{i2} = 0V$



$V_{o1} = -\frac{R_2}{R_1} V_{i1}$

in V_o

$\therefore$ Total output $V_o = V_{o1} + V_{o2} = -\frac{R_2}{R_1} V_{i1} + \frac{R_4}{R_3+R_4}\left(1+\frac{R_2}{R_1}\right) V_{i2}$

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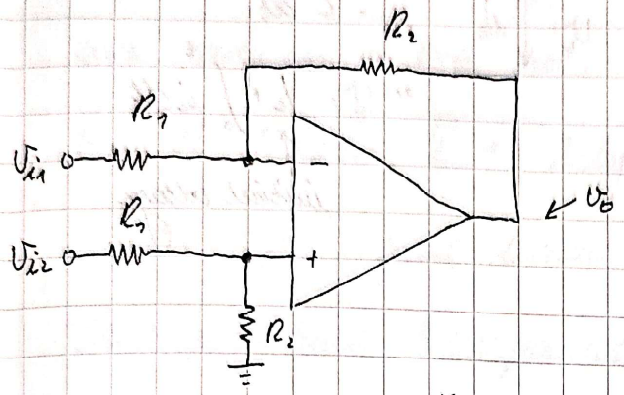
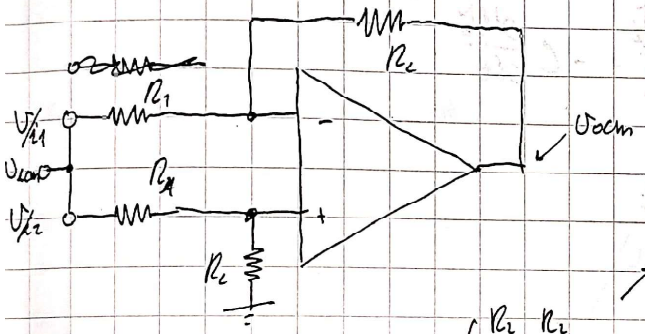
Let $R_3 = R_1$, $R_4 = R_2$



$$v_o = \cdots = \frac{R_2}{R_1}(v_{i2} - v_{i1}) = \frac{R_2}{R_1} v_{id}$$

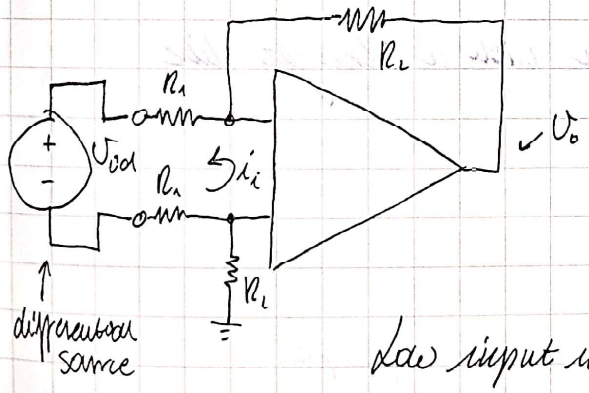
For common mode:



$$v_{ocm} = \left(\frac{R_2}{R_1} \frac{R_2}{R_1 + R_2} - \frac{R_2}{R_1} + \frac{R_2}{R_1 + R_2} \right) v_{icm} = 0$$

(the bracketed term $\to 0$)

Input resistance of a differential amplifier



KVL: $v_{id} = R_1 i_i + 0 + R_1 i_i$

$\Rightarrow v_{id} = 2R_1 i_i$

$\therefore R_{id} = 2R_1$

Low input impedance, when needed, use an instrumentation amplifier.

Quiz 2

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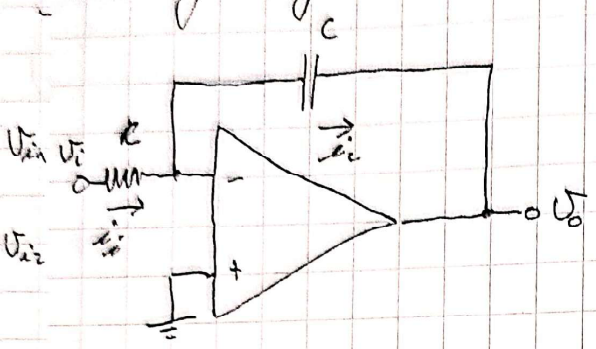
Inverting integrator



$i_c = C \frac{dV_c}{dt}$

$\Rightarrow V_c = V_c + \int_0^t i_c dt$

↑ initial voltage

$V_+ = V_-$, $V_+ = 0V \Rightarrow V_- = 0V$

$i_i = i_- = 0A \Rightarrow i_i = i_c$

$i_i = \frac{V_i - V_-}{R} = \frac{V_i}{R}$

$i_c = C \frac{d}{dt}(V_- - V_o) = -C \frac{dV_o}{dt}$ $\Bigg\} \Rightarrow \frac{V_i}{R} = -C \frac{dV_o}{dt}$

$\therefore V_o = -V_c - \frac{1}{RC} \int_0^t V_i dt$

V_{in}

→ can be used as an LPF

Introduction to Bode plots

Bode plots are a fast and easy method to approximate the frequency response of a transfer function with a straight line approximation on a log-log scale.

given a generic transfer function

$$T(j\omega) = \frac{K\left(1 + \frac{j\omega}{z_1}\right)}{j\omega\left(1 + \frac{j\omega}{p_1}\right)}$$

K: constant, z_1: zero, p_1: pole

zeros on top

poles on the bottom

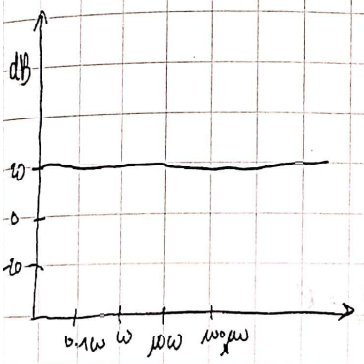
$$|T(j\omega)| = |K| \frac{|1 + \frac{j\omega}{z_1}|}{\omega |1 + \frac{j\omega}{p_1}|}$$ take log of each

each contributes to Bode plot:

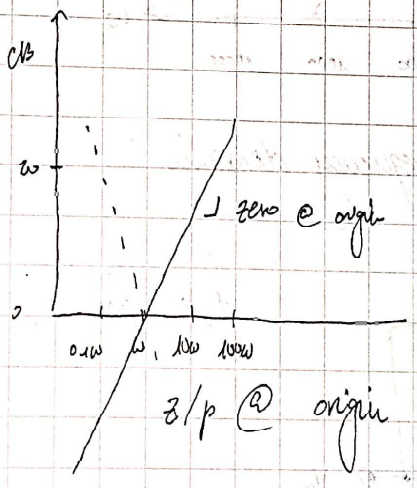
$$20\log|K| + 20\log|1 + \frac{j\omega}{z_1}| - 20\log B_{ac}(\omega) - 20\log|1 + \frac{j\omega}{p_1}|$$

→ add tops, subtract bottoms

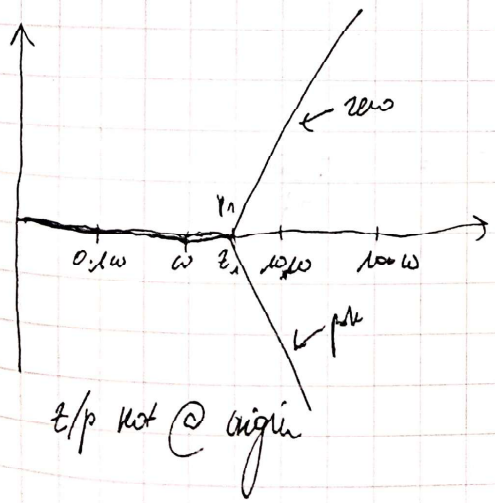
Bode plots have 3 components: constant (k), poles or zeros at origin $(j\omega)$, poles or zeros not at origin $(1 + \frac{j\omega}{z_1})$, $(1 + \frac{j\omega}{p_1})$



K



z/p @ origin



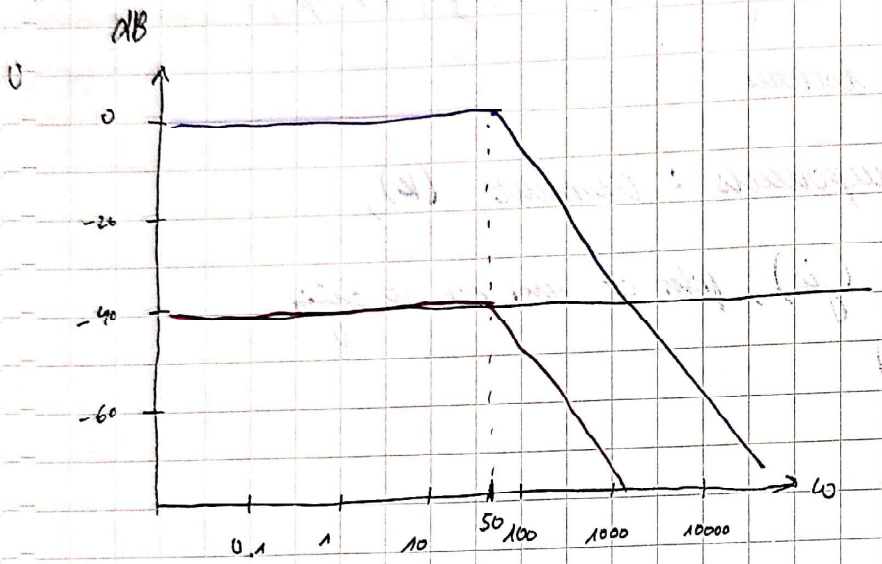
z/p not @ origin

Bode plot: LPF example

$$T(j\omega) = \frac{\frac{1}{100}}{\frac{100}{100} + j\frac{\omega}{100}} = \frac{\frac{1}{100}}{1 + j\frac{\omega}{50}}$$

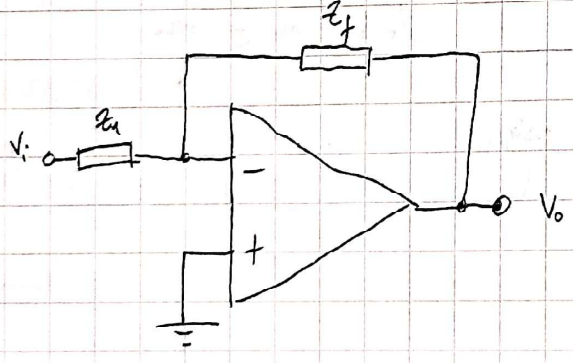
$K = \frac{1}{100}$

$P_1 = 50$



- $K: 20\log\left(\frac{1}{100}\right) = -40$
- $P: 20\log\left|\frac{1}{1+j\frac{\omega}{P_1}}\right|$
- Total

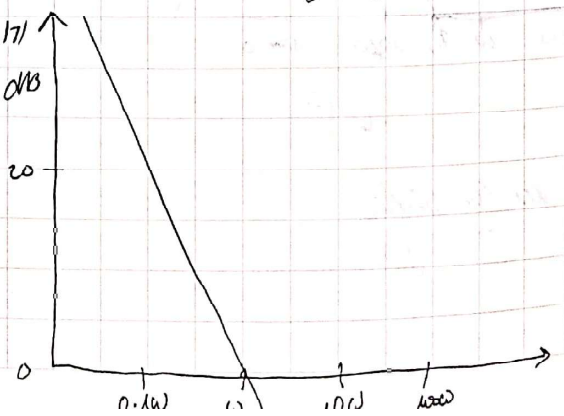
Inverting integrator in frequency domain



$Z_f = \frac{1}{j\omega C}$, $Z_1 = R$

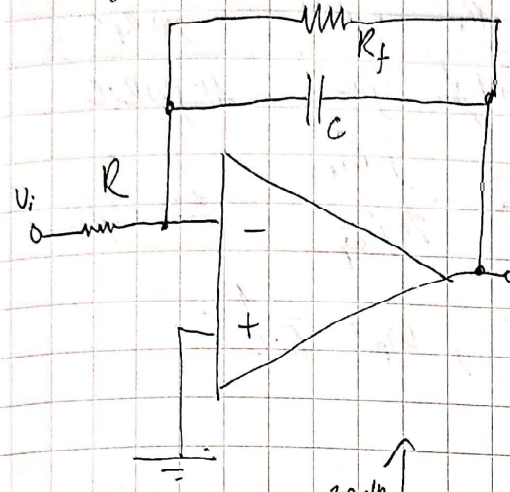
$$T = \frac{V_o}{V_i} = -\frac{Z_f}{Z_1} = -\frac{1}{j\omega RC} = -\frac{1}{j\omega\tau} = -\frac{\frac{1}{\omega}}{j\frac{\omega}{\omega_0}}$$

$\tau = RC \Rightarrow \omega_0 = \frac{1}{RC}$



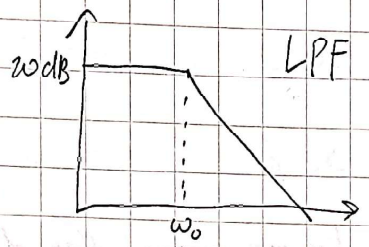
super high gain @ DC

Integrator with a parallel feedback resistor



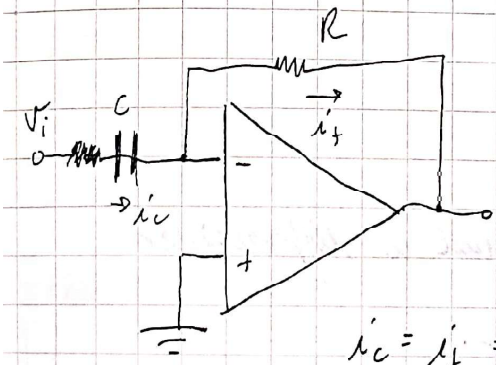
$$Z_f = R_f \parallel Z_c = \frac{R_f}{1+j\omega R_f C}$$

$$T(j\omega) = \frac{V_o}{V_i} = -\frac{Z_f}{Z_i} = \frac{-\frac{R_f}{R}}{1+j\omega R_f C}$$



LPF response

Op amp differentiator



$$i_f = \frac{V_- - V_o}{R} = -\frac{V_o}{R}$$

$$i_c = C\frac{dV_c}{dt}$$

$$i_c = i_f \Rightarrow V_o = -RC\frac{dV_c}{dt} \qquad V_i = V_c$$

$$\therefore V_o = -RC\frac{dV_i}{dt}$$

In frequency domain:

$$\frac{V_o}{V_i} = -\frac{Z_f}{Z} = -\frac{R}{\frac{1}{j\omega C}} = -j\omega RC = T(j\omega)$$

$|T|\big|_{\omega \to \infty} = \infty$, $|T|\big|_{\omega \to 0} = 0$ high pass filter

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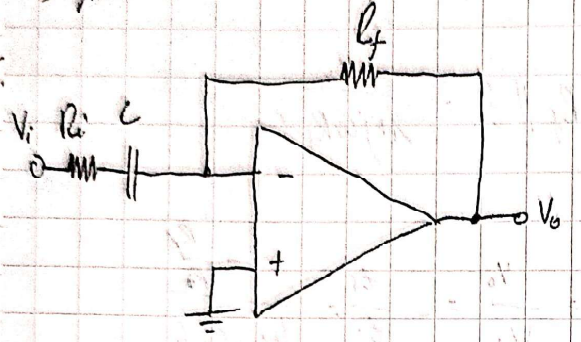
Differentiator with a series input resistor:

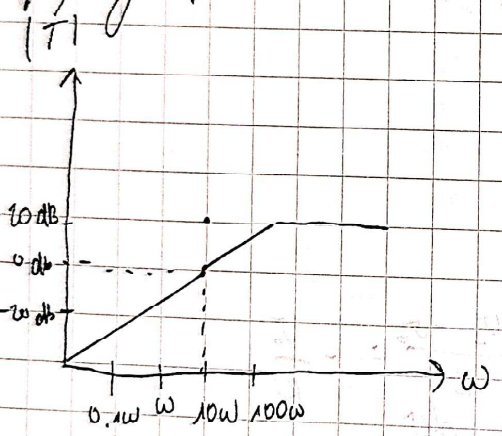


$$\frac{V_o}{V_i} = -\frac{R_f}{R_i + \frac{1}{j\omega C}} \cdot \frac{j\omega C}{j\omega C} = -\frac{j\omega R_f C}{1 + j\omega R_i C}$$

$$= -\frac{j\frac{\omega}{\omega_z}}{1 + j\frac{\omega}{\omega_p}} \leftarrow T(j\omega)$$

$$\omega_z = \frac{1}{R_f C}, \quad \omega_p = \frac{1}{R_i C}$$

frequency response



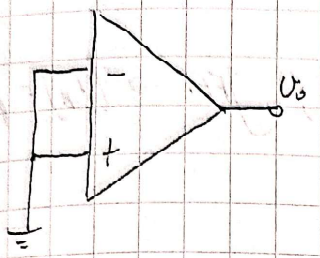
frequency response that of a high pass filter

→ A series combination of an integrator and a differentiator can be used to form a bandpass filter.

Date

Imperfections of an op amp at DC

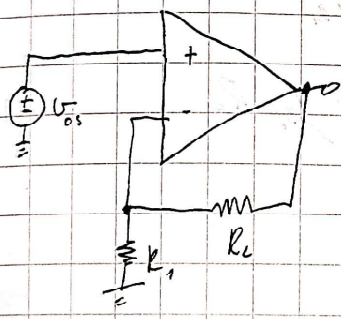
1) Offset voltage (V_{os})



$$V_o = A(V_+ - V_-) = \pm V_{limit} \neq 0$$

↑ saturation limit set by external supply rails

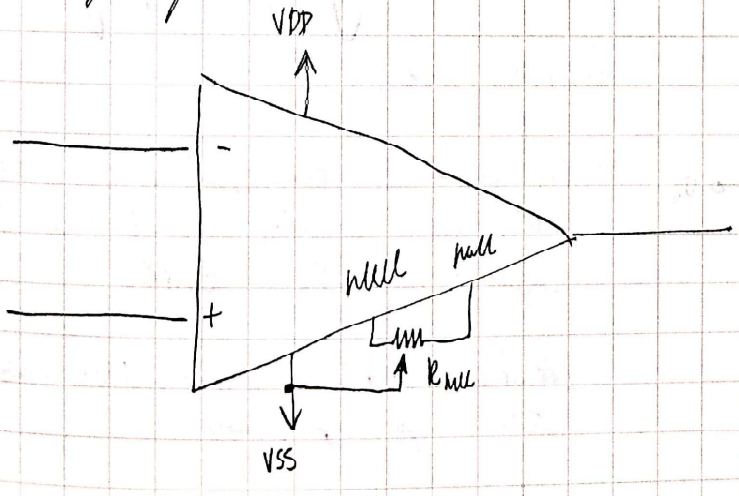
This happens because the input transistors (BJTs or FETs) aren't matched perfectly. V_{os} is usually between 1 and 5mV.



$$V_{os} = V_{os}\left(1 + \frac{R_i}{R_1}\right)$$

This offset is amplified and needs to be taken care of when dealing with small low amplitude input signals.

This can be dealt with using an op amp with an offset null point adjustment.



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2) Input bias current and offset:

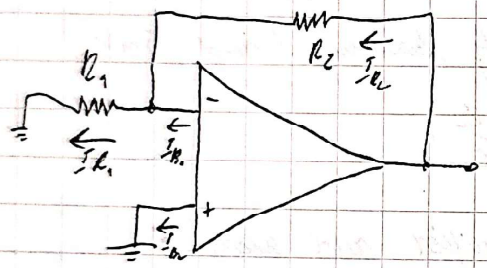
→ Input bias current: $I_B = \frac{I_{B_1} + I_{B_2}}{2}$

→ Input offset current: $I_{os} = |I_{B_1} - I_{B_2}|$

Typical input bias current: BJT ↔ $[I_B]$ = nA, JFET → $[I_B]$ = pA

I_{os} is usually 1/10 of input bias current.

Impact of input bias current



$I_{R_1} = \frac{V_0 - 0}{R_1} = 0$

$I_{R_2} = I_{R_1} + I_{B_1} = I_{B_1}$

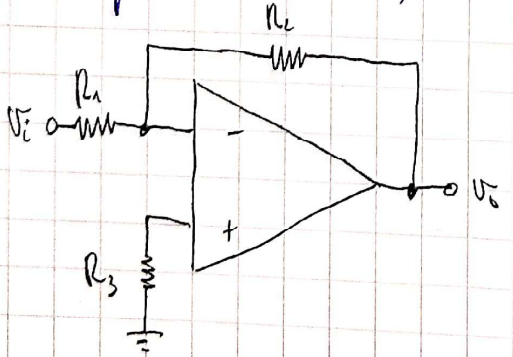
$V_0 = V_{R_2} = I_{R_2} \cdot R_2 = I_{B_1} \cdot R_2$

If I_{B_1} = 100nA and R_2 = 1 MΩ

$V_0 = 10^{-7} \cdot 10^6 = 100\,mV$

How to deal with input bias current?

↳ place a resistor between the non-inverting input and ground

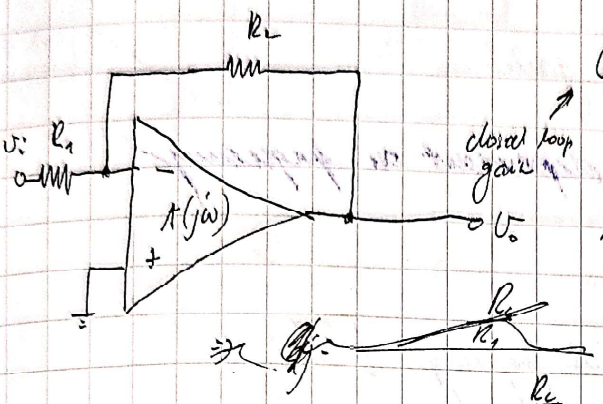


Rules of thumb:

- R_3 should be the equivalent thevenin resistance looking out of the inverting input; here $R_3 = R_1 \| R_2$
- don't use resistance values higher than 500kΩ for op amps

Date

Frequency response of a closed loop op amp



$$G = \frac{v_o}{v_i} = \frac{\frac{R_2}{R_1}}{1 + \frac{1}{A}\left(1 + \frac{R_2}{R_1}\right)}$$

let $A(j\omega) = \frac{A_0}{1 + j\frac{\omega}{\omega_b}}$ (LPF) ← B

$$\Rightarrow G(j\omega) = \frac{\frac{R_2}{R_1}}{1 + \frac{1 + \frac{R_2}{R_1}}{\frac{A_0}{1 + j\frac{\omega}{\omega_b}}}} = \cdots = \frac{\frac{R_2}{R_1}}{1 + \frac{1}{A_0}\left(1 + \frac{R_2}{R_1}\right) + \frac{j\omega}{\omega_t}\frac{1}{1 + \frac{R_2}{R_1}}}$$

$\omega_t = A_0 \omega_b$

Op amps don't work at really high frequencies due to their LPF characteristic. For high frequency applications, use dedicated op amps rated for that frequency or use a dedicated circuit with that frequency in mind (looking at you, RF). General purpose op amps generally operate at a max of 10s ~~to~~ to 100s of kHz. Specialized op amps like MAX4395 (video op amp) will happily run at 10s of MHz.

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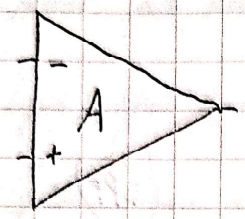
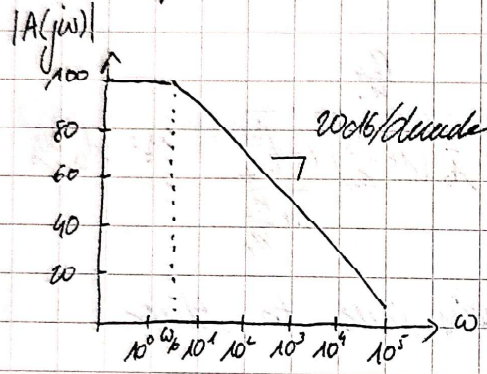
Finite open loop gain and bandwidth



A - suggests finite gain

In reality, gain is dependent on frequency

Bode plot:



ω_b : break frequency $\left[\frac{rad}{s}\right]$

20dB/decade $f_b = \frac{\omega_b}{2\pi}$ also break frequency

ω_t, f_t : unity gain frequency

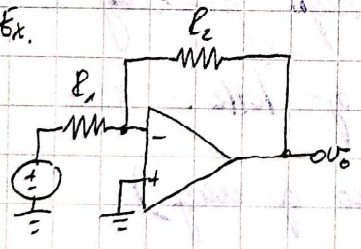
$$A(j\omega) = \frac{A_0}{1 + j\frac{\omega}{\omega_b}}$$ A_0 : DC gain

→ gain falloff begins at ω_b

→ for $\omega \gg \omega_b$, $A(j\omega) \approx \frac{A_0 \omega_b}{j\omega}$

$$\rightarrow \omega_t = \left.\frac{A_0 \omega_b}{j\omega}\right|_{\omega=\omega_t} = 1 \Rightarrow \omega_t = A_0 \omega_b$$

Ex.



$$G = \frac{v_o}{v_i} = -\frac{\frac{R_2}{R_1}}{1 + \frac{1}{A}\left(1 + \frac{R_2}{R_1}\right)} \Rightarrow G = -\frac{\frac{R_2}{R_1}}{1 + \frac{1}{A_0}\left(1 + \frac{R_2}{R_1}\right) + \frac{j\omega}{\omega_t \frac{1}{1 + \frac{R_2}{R_1}}}}$$

$$\Rightarrow G \simeq -\frac{\frac{R_2}{R_1}}{1 + \frac{j\omega}{\frac{\omega_t}{1 + \frac{R_2}{R_1}}}}$$ where $\omega_0 = \frac{\omega_t}{1 + \frac{R_2}{R_1}}$ ← break frequency

When $A_0 \gg 1 + \frac{R_2}{R_1}$

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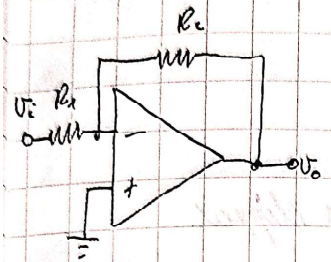
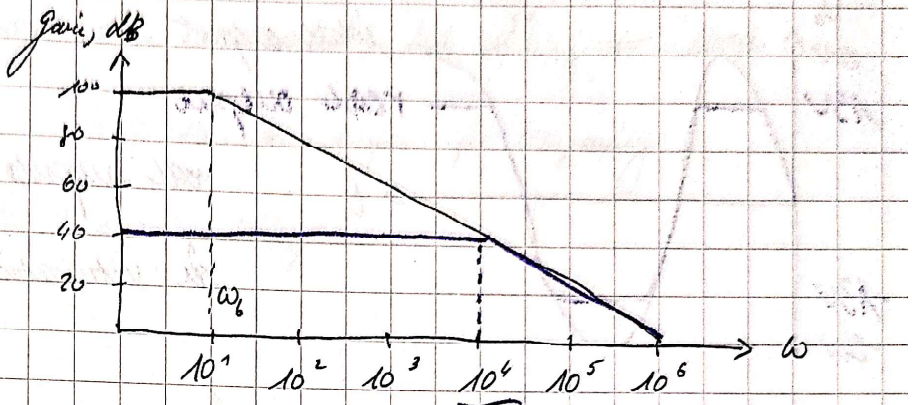
Ex. $A_0 = 10^5 = 100$ dB, $\frac{R_2}{R_1} = 100$, $\omega_b = 10 \frac{rad}{s}$





$$G = -\frac{\frac{R_2}{R_1}}{1 + \frac{j\omega}{\frac{\omega_b}{1 + \frac{R_2}{R_1}}}}$$

$$\omega_t = A_0 \omega_b = 10^5 \cdot 10 = 10^6 \frac{rad}{s}$$

$$\omega_0 = \frac{\omega_t}{1 + \frac{R_2}{R_1}} = \frac{10^6}{1 + 100} \simeq 10^6 \cdot \frac{1}{100} = 10^4 \frac{rad}{s}$$ ω_0 : cutoff

→ ω_0 or f_0 is the maximum frequency we can reliably operate the op amp at and below.

Large signal operation of an op amp

Another non-optimal characteristic of an op amp is output voltage saturation. The maximum output voltage of an op amp is usually a few volts below its supply voltage.

Typically: $V_{o_{max}} = +V_S - 2V$ and $V_{o_{min}} = -V_S + 2V$

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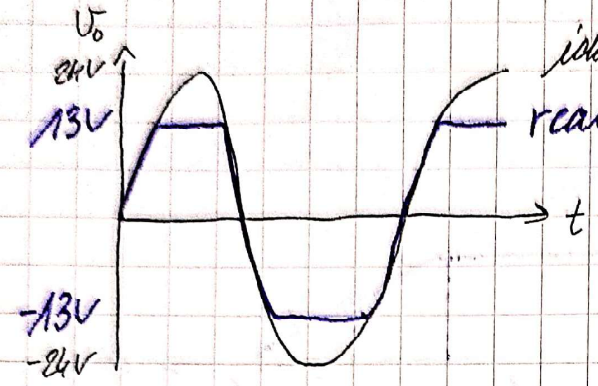
Example at $+V_s = 15V$ and $-V_s = -15V$



real output gets clipped in saturation

Output current limits: op amps usually can't supply much current at their output. This value is device-specific and usually between 10 and 100 mA. Be careful when driving low-impedance loads.

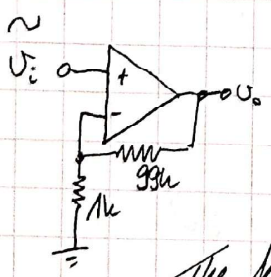
Ex. $I_{o\,max} = 50\,mA$, $R_L = 8\Omega$

$\Rightarrow U_{max} = I_{o\,max} \cdot R_L = 50mA \cdot 8\Omega = 400\,mV$ pk-pk

this value doesn't change regardless of input voltage or gain

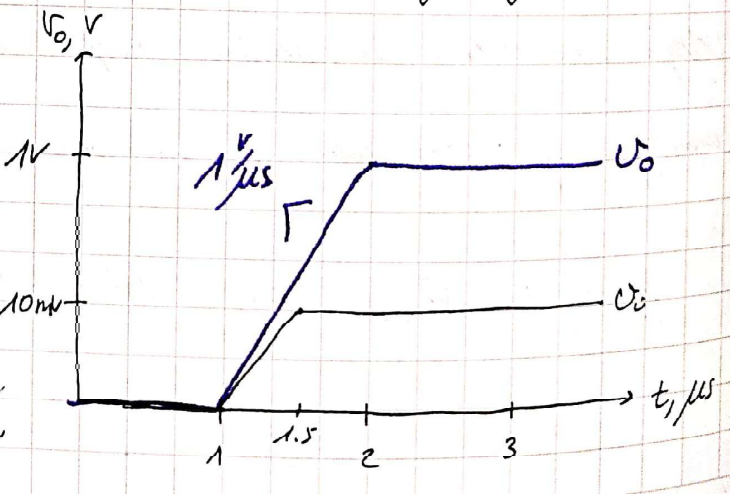
Slew rate is the maximum rate at which the output voltage can change. It's usually device-specific (see datasheet). Typically 1 to 5 $\frac{V}{\mu s}$

Ex.



$U_o = 100\,U_i$

$SR = 1\frac{V}{\mu s}$



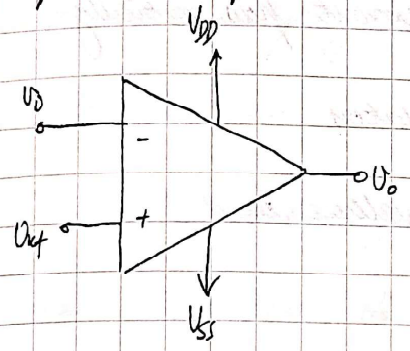
The faster the transition, the more high-frequency content in the signal.

Date

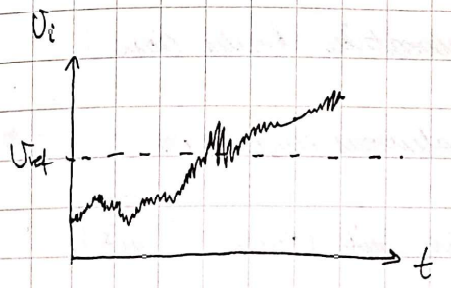
Comparators

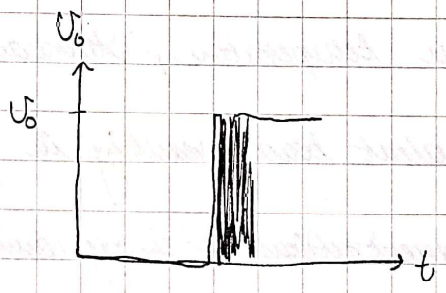
An op amp can be used as a comparator, but it's slower than using a dedicated IC. Comparators are useful for comparing time-varying signals to a reference voltage.

Op amp as a comparator:

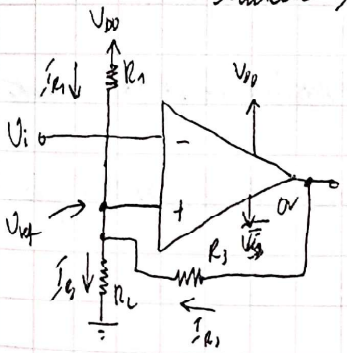


$$\begin{cases} V_i > V_{ref} \Rightarrow V_o = V_{SS} \\ V_i < V_{ref} \Rightarrow V_o = V_{DD} \end{cases}$$





↳ Solution to noise: hysteresis (Schmitt trigger)

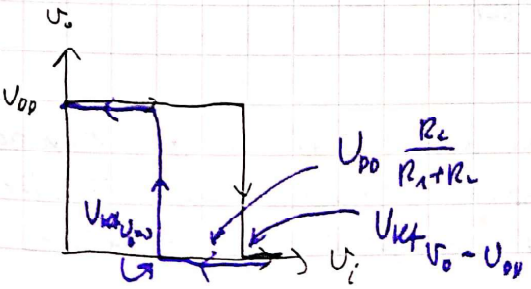


$V_o = V_{DD}$

$$I_{R_1} = \frac{V_{DD} - V_{ref}}{R_1} \wedge I_{R_2} = \frac{V_{ref}}{R_2} \wedge I_{R_3} = \frac{V_{DD} - V_{ref}}{R_3}$$

$$\Rightarrow \frac{V_{ref}}{R_2} = \frac{V_{DD} - V_{ref}}{R_1} + \frac{V_{DD} - V_{ref}}{R_3}$$

$$\Rightarrow \dots \Rightarrow V_{ref} = V_{DD} \frac{R_1 R_2 + R_2 R_3}{R_1 R_3 + R_1 R_2 + R_2 R_3}$$



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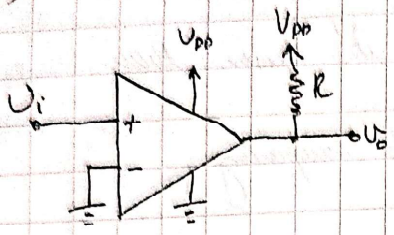
Idealized comparator: open-collect output, high slew rate ($1\frac{V}{ns}$)



$U_i < 0 \Rightarrow U_o = V_{DD}$

$U_i > 0 \Rightarrow U_o = 0$

Semiconductor and a PN junction (slides)

→ Semiconductors: materials in which current flows partially as compared to conductors and insulators

→ Silicon: group IV, 4 electrons in outer valence band

→ Intrinsic silicon:

- at room temperature, thermal generation breaks down some covalent bonds resulting in free electrons and holes with concentrations of charge carriers per unit volume $\left[\frac{n}{cm^3}\right]$
- some free electrons recombine with bound holes: recombination
- recombination results in the appearance of moving holes: hole current
- recombination rate is proportional to temperature, # of holes and free e^-
- density of electrons or holes in intrinsic silicon:

$$n_i = BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}}$$

n_i: number of electrons or holes in pure or intrinsic silicon at room temperature
T: temperature, B: material dependent, E_g: bandgap energy (Si: 1.12 eV)
k: Boltzmann constant; $n_i = 1.5 \cdot 10^{10} \frac{N}{cm^3}$ for Si at room temperature

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- in thermal equilibrium, # of free electrons and holes is equal

$$n = p = n_i \quad <=> \quad n \cdot p = n_i^2$$

→ N-type doped silicon

- added impurity from group V, usually phosphorus
- donor concentration $N_D \gg n_i$
- free electrons (majority carriers): $n_n \approx N_D$

→ P-type doped silicon

- added impurity from group III, usually boron
- donor concentration $N_A \gg n_i$
- free hole (majority carrier) concentration: $p_p \approx N_A$

⇒ Drift current

- drift current, comprised of hole and electron flows, occurs due to the force of an electric field
- hole drift velocity: $v_{p\,drift} = \mu_p \cdot E$

  μ_p: hole mobility $\left[\frac{cm^2}{Vs}\right]$, E: electric field $\left[\frac{V}{cm}\right]$
- electron drift velocity: $v_{n\,drift} = \mu_n E$

$$\mu_n = 1350 \frac{cm^2}{Vs}, \quad \mu_p = 480 \frac{cm^2}{Vs} \qquad \mu_n > \mu_p$$

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- hole drift current : $I_p = Aqp v_{p\,drift}$ $\left[\frac{C}{s} = A\right]$

  A : area $[cm^2]$, q : electron charge $[C]$, p : hole carrier concentration $\left[\frac{1}{cm^3}\right]$

  $I_p = Aqp\mu_p E$ $\quad J_p = \frac{I_p}{A}$ $\quad J_p$: hole current density

- electron drift current : $I_n = -Aqn\mu_n E$

- total drift current density : $J = J_p + J_n = q(p\mu_p + n\mu_n)E$

→ Diffusion current : occurs when carrier distribution in the bulk material isn't uniform

→ Open circuit PN junction :

- $V_T = \frac{kT}{q}$: thermal voltage, 25.9mV at room temp (295K)

→ PN junction in forward bias

- applied voltage subtracts from the built-in voltage which allows more majority carriers to diffuse across the junction
- the depletion region decreases to support the lower voltage, this causes the total current to become larger

$$I_{total} = I_D - I_S \;,\; I_D \gg I_S$$

→ PN junction U-I characteristic

$$I = I_S\left(e^{\frac{V}{V_T}} - 1\right)$$

$$I_S = Aqn_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right)$$

L_n : diffusion length in n material

L_p : diffusion length in p material

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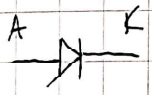
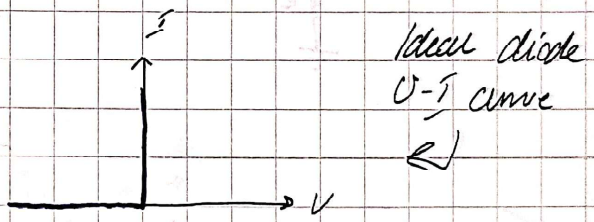
⇒ Reverse breakdown of a PN junction:

- higher reverse voltage can't be applied and $I = -I_S$ can't be maintained
- at $U = U_Z$) a high reverse current will flow (can be destructive if not limited)
- Zener breakdown: increased E-field breaks apart covalent bonds ~~electrons~~ resulting in many more minority carriers
- Avalanche breakdown: minority carriers accelerate, fast and break down to covalent bonds releasing charges... cycle repeats

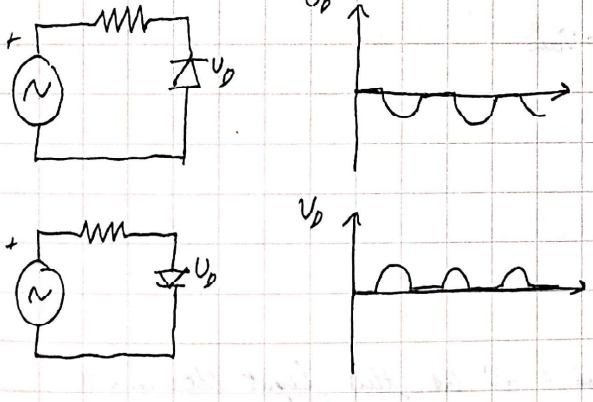
Silicon / PN junction diodes



nonlinear device



Example of operation:



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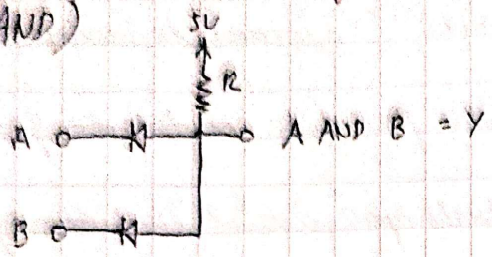
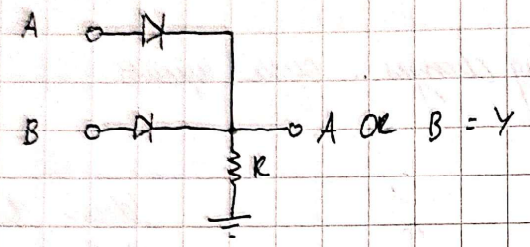
The diode as a logic operator

AND)



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

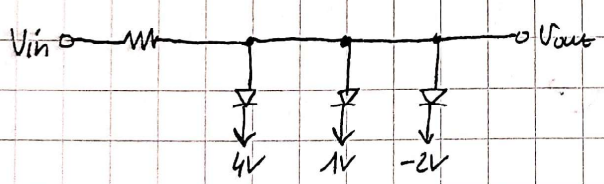
OR)



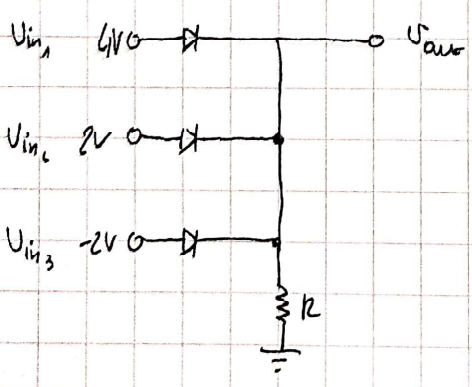
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

The diode as a minimum and maximum detector

MIN)

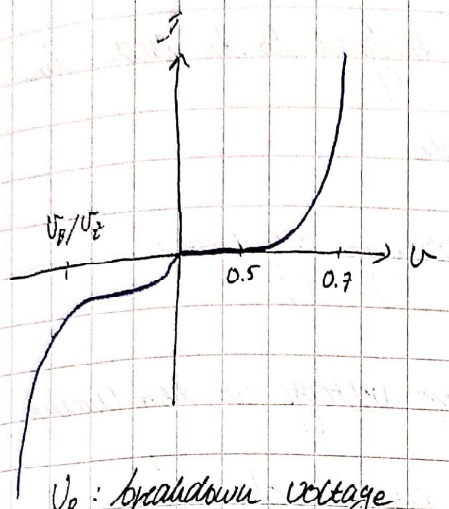


MAX)



V_{out} = 4V as this input dominates and reverse biases other diodes

Characteristics of PN junction diodes



- in forward bias: $(V > 0)$

$$i = i_S \left(e^{\frac{v}{V_T}} - 1 \right) \quad [A]$$

$$i_S = A q n_i^2 \left(\frac{D_P}{L_P N_D} + \frac{D_N}{L_N N_N} \right)$$

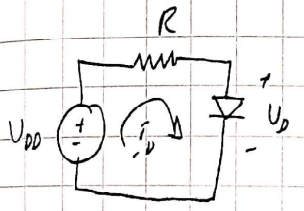
- in reverse bias

$$i \simeq i_S$$

V_B : breakdown voltage

V_Z : Zener voltage

Forward bias characteristic model:



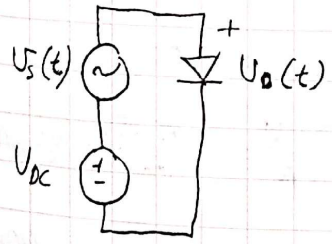
$V_{DD} > 0.5V$ $\quad i_D = i_S e^{\frac{V_{DD}}{V_T}}$

$$V_{DD} = i_D R + V_D$$

$$i_D = \frac{V_{DD} - V_D}{R}$$

Small signal model of a diode !

→ The voltage across the diode has a DC and an AC component.



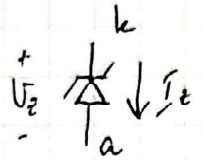
$$v_D(t) = v_s(t) + V_{DC}$$

small signal resistance approximation: $r_d = \frac{V_T}{I_D}$

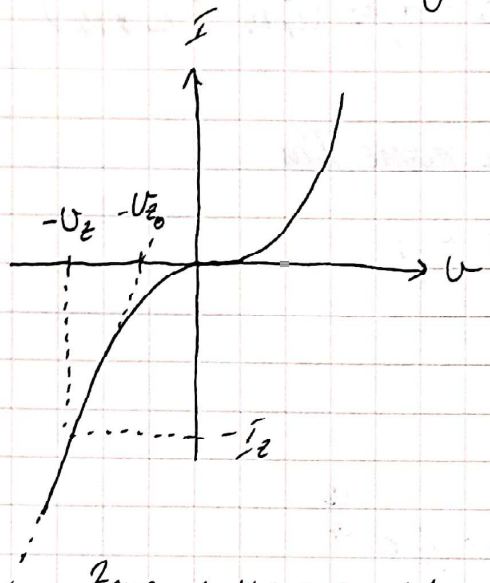
small signal V-i characteristic: $i_d(t) = \frac{I_D}{V_T} v_d(t)$

v_d

Zener diodes



- high impedance devices designed to be used in reverse breakdown mode
- commonly used in voltage regulators

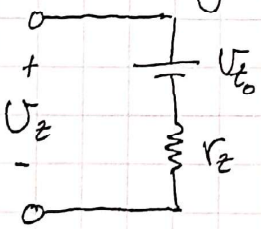


V_{z_0} : zener voltage at test current I_z

V_{z_0} : zero crossing of $\frac{1}{r_z}$

$\frac{1}{r_z}$: straight line approximation when the zener diode is conducting

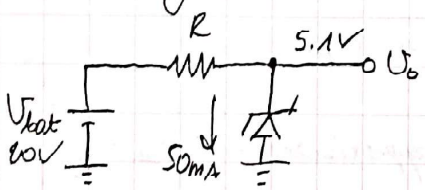
Zener voltage model



$V_z > V_{z_0}$

$V_z = V_{z_0} + r_z I_z$

Ex. Design a 5.1 zener-regulated PSU ($r_z = 7\Omega$ @ 50mA)



$R = \frac{20 - 5.1}{50mA} = 298\Omega \approx 300\Omega$

V_{bat} drops to 15V: $I = \frac{V}{R} = \frac{15 - 5.1}{300} = 33mA$

$\Delta I_z = I_{z_{20V}} - I_{z_{15V}} = 17mA \Rightarrow \Delta V_{out} = r_z \cdot \Delta I_z = 7\Omega \cdot 17mA = 120mV$

$\Rightarrow V_{out} = 5.1V - 120mV = 4.98V$

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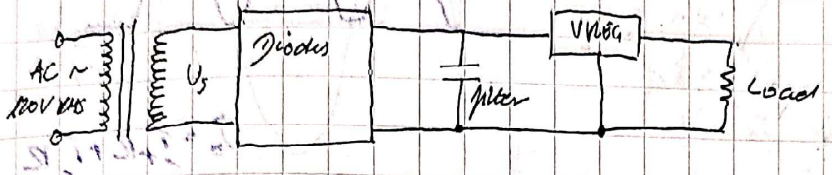
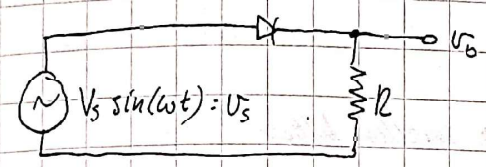
Rectifiers

LOL, I WROTE THIS SECTION DRUNK AF SWEDISH WOMEN DOES THE JOB

→ An important application using diodes to convert AC to DC

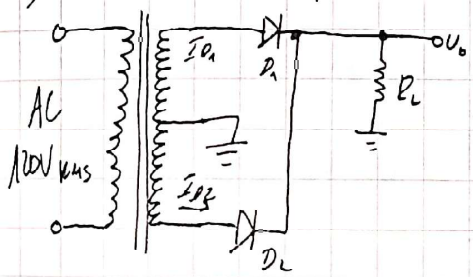


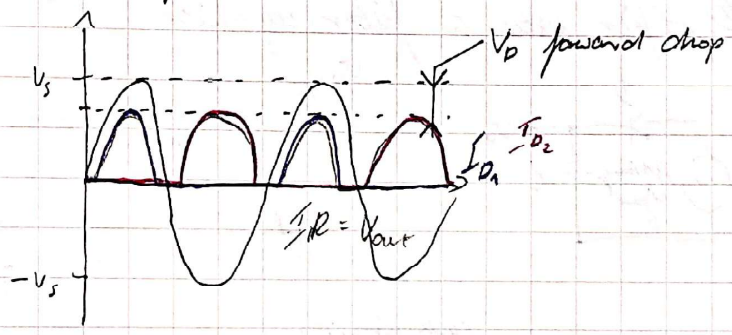
1) Half wave rectifier



- For $U_s < U_D$, no current flows (U_s less than forward voltage)
- Max reverse voltage across the diode is $-V_s$, diode specified at 150 % V_s

2) Full wave rectifier - center tap transformer





$$V_{rev_{max}} = U_{o_{max}} + U_s \simeq 2V_s$$

→ 2 diodes have to block $2V_s$ in reverse

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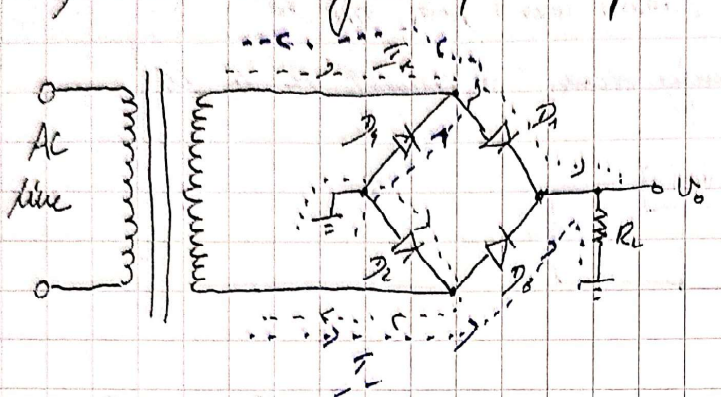
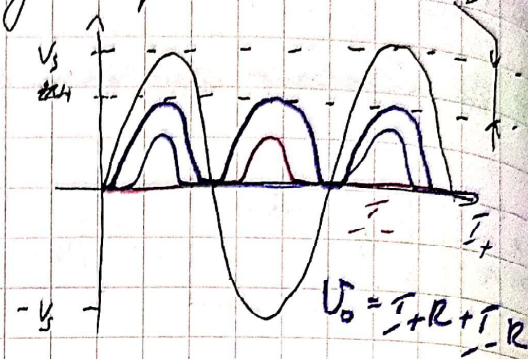
3) Full wave bridge rectifier (full bridge rectifier)





- Current flows in the same direction through R
- Max reverse voltage : $V_{R_{max}} = V_s - V_o$

\+ • Uses positive and negative half waves → smaller filter

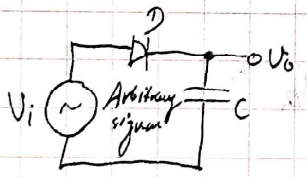
\+ • Simpler and cheaper transformer

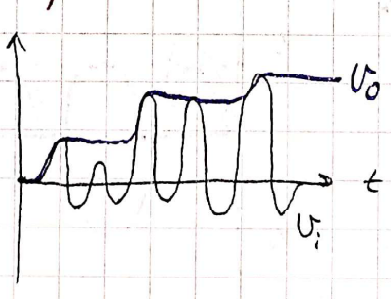
\+ • Smaller max diode reverse voltage

\− • Uses more diodes but they're cheap now

+ : advantages − : disadvantages

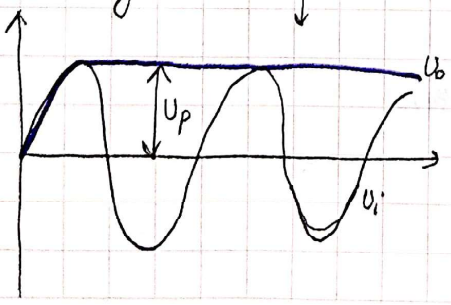
Rectifier with a filter capacitor





Sine signal

open circuit



Once $V_o = V_p$, the cap stays charged

V_p : peak input, assuming ideal diode

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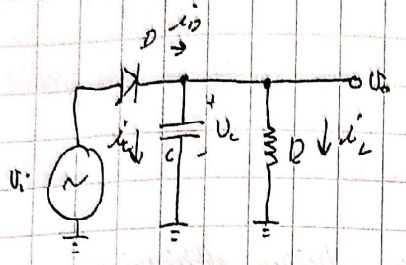
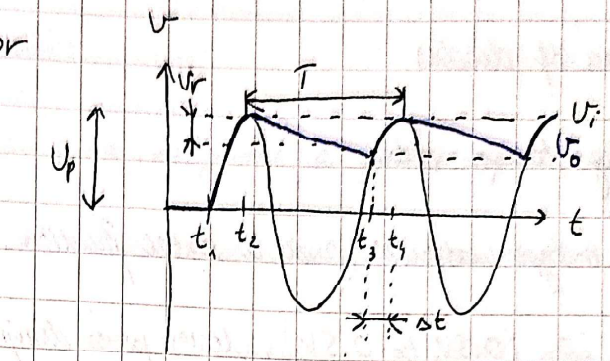
Now with a load resistor





t_1: V_o follows V_i up to V_p, capacitor charges

t_2: V_i drops and D opens since $V_c > V_i$, C starts discharging thru R

t_3: V_c falls ~~until~~ until equal to V_i which turns the diode back on and C starts to charge again

t_4: cycle repeats

Find ripple voltage V_r in terms of V_p, T, R, C

Assume $RC \gg T$, $i_L = \frac{V_o}{R}$, $i_D = i_c + i_L = C\frac{dV_i}{dt} + \frac{V_o}{R}$ (when diode on)

Let $\bar{i}_L$ be the DC component of i_L or its average

$\bar{i}_L \approx \frac{V_p}{R}$ since $RC \gg T$ & $V_r \ll V_p$

When diode is off: $t \in [t_2, t_3]$: $V_o = V_p e^{-\frac{t}{RC}}$

@t_3: $V_p - V_r = V_p e^{-\frac{t}{RC}} \simeq V_p\left(1 - \frac{T}{RC}\right) \simeq V_p \frac{T}{RC}$

$e^{-x} \simeq 1 - x$ for small x

@Δt: $V_p \cos(\omega \Delta t) = V_p - V_r$ where $\omega = 2\pi f = \frac{2\pi}{T}$

$\omega\Delta t$ is a small angle: $\cos(\omega\Delta t) \simeq 1 - \frac{1}{2}(\omega\Delta t)^2$ $\quad V_r \simeq V_p \frac{T}{RC}$

$V_p\left(1 - \frac{1}{2}(\omega\Delta t)^2\right) \simeq V_p - V_r \Rightarrow \omega\Delta t \simeq \sqrt{\frac{2V_r}{V_p}} \Rightarrow$

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Other types of diodes

1) Schottky barrier diode

⇒ only n-type material and a metal junction

\+ low U_f (0.3V to 0.5V), lower power dissipation, higher efficiency

\+ no stored charge effect ⇒ high speed (RF, clamping)

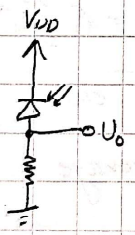
\- low reverse voltage

\- high reverse leakage current

2) Photodiodes

→ PN junction with a really large depletion region exposed to light

→ can act as a current source or a variable resistor



U_o to detector; high gain needed due to a very low current output (μA)

3) LED

→ emits light when driven in the forward direction

→ light color depends on the semiconductor used

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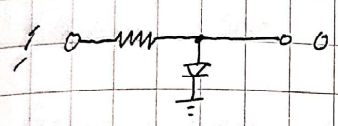
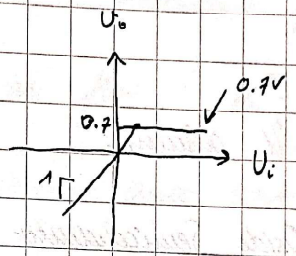
Clamping diodes as limiters

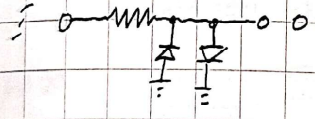
There are instances where you want to protect a portion of a circuit from input voltages large enough to cause damage. A solution is including clamping diodes at the circuit's frontend to limit the voltage.

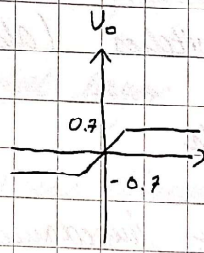
1) Single diode limiter





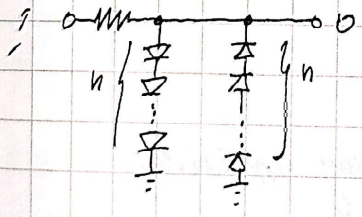
2) Two diode limiter

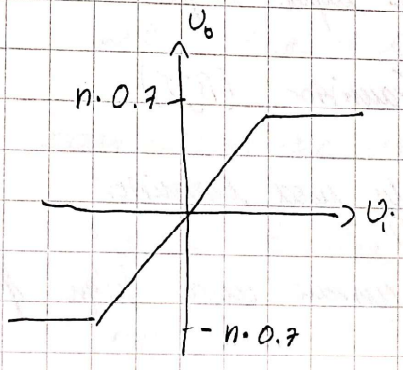




clamps between -0.7 V and 0.7V

3) Stacked diode limiter





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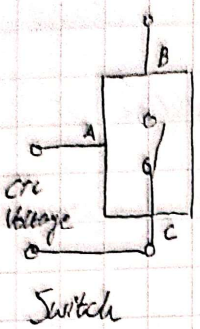
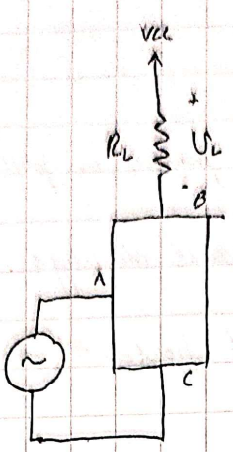
Semiconductor



Switch



amplifier

2 types of transistors we'll discuss:

- Enhancement Metal Oxide Semiconductor Field Effect Transistor (MOSFET)
  - used in ICs as switches (also JFETs)
  - can be made very small
  - used where a high input impedance is needed
- Bipolar Junction Transistor (BJT)
  - the first commonly used transistor
  - used in high current and high frequency applications

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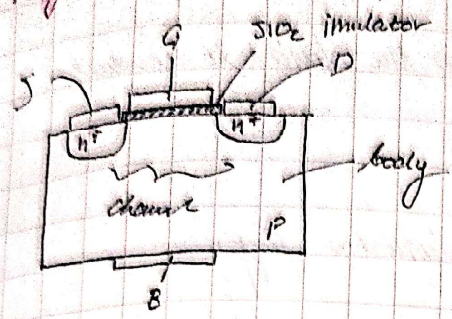
Physical operation of an enhancement mode MOSFET



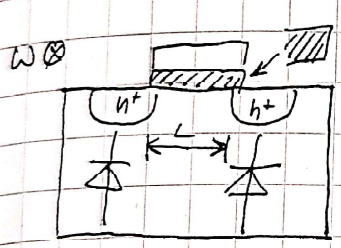
Body: p-type silicon substrate

Source: n^+-type; supplies the electrons that will make up the output current flow

Drain: where the electrons flow to n^+-type

Channel: region between drain and source

Gate: located above channel, formed by a silicon dioxide insulator and a metal layer



typical dimensions: $L \simeq 0.03\,\mu m$,

$W \simeq 0.1\,\mu m \rightarrow 100\,\mu m$, $t_{ox} = 1 \rightarrow 10\,nm$

body diodes are usually reverse biased

$v_{SB} > 0$ & $v_{DB} > 0$

→ body shorted to S and $v_B > v_S$

→ zero gate voltage: $v_{GS} = 0V$, off-state (cutoff)

→ Positive v_{GS} ($v_{GS} > 0$): depletion region forms and electrons can move more freely from S to D. Holes are pushed deeper into the substrate

Definitions:

- V_t : threshold voltage ; minimum V_{GS} for the device to conduct
- V_{ov} : overdrive voltage ; $V_{ov} = V_{GS} - V_t$
- C_{ox} : capacitance across gate and SiO_2 insulator layer
  - ↳ positive charge at the gate, negative charge at the channel

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.9 \cdot \varepsilon_0}{t_{ox}} \quad \left[\frac{F}{m^2}\right]$$

  - ↳ total stored charge : $|Q| = C_{ox} \cdot W \cdot L \cdot V_{ov}$

Let $V_{GS} > V_t$ and very small V_{DS}

- current begins to flow, $i_D > 0$
- expression for i_D :
  - charge per unit length : $\frac{Q}{L} = C_{ox} \cdot W \cdot V_{ov}$
  - electric field across the channel : $E = \frac{V_{DS}}{L}$
  - electron drift velocity : $v = \mu_n E = \mu_n \frac{V_{DS}}{L}$
  - actually i_D :

$$i_D = \frac{Q}{L} v = C_{ox} W V_{ov} \frac{\mu_n V_{DS}}{L} = \underbrace{C_{ox} \frac{W}{L} \mu_n (V_{GS} - V_t)}_{\text{conductance } g_{DS} = \frac{1}{r_{DS}}} V_{DS}$$

$$g_{DS} = \underbrace{C_{ox} \frac{W}{L} \mu_n}_{\text{physical param}} \underbrace{(V_{GS} - V_t)}_{V_{ov}} = \underbrace{k_n}_{\text{transconductance}} (V_{GS} - V_t)$$

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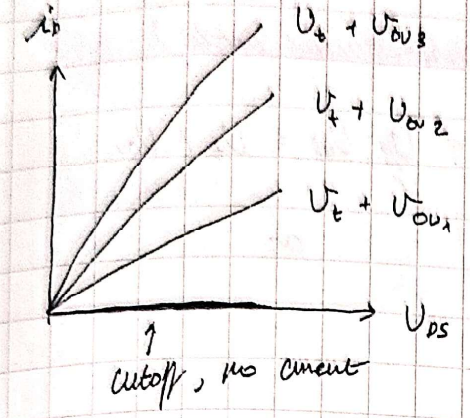
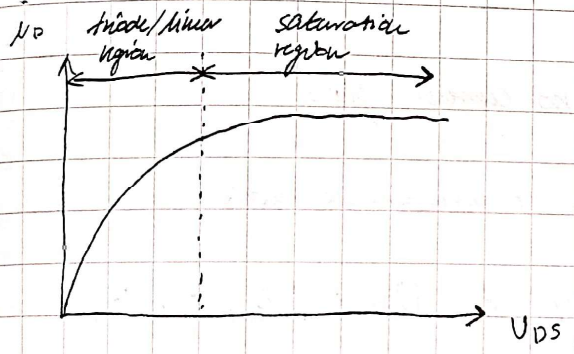


Straight lines indicate that the MOSFET acts as a resistor for small V_{DS}

Let V_{GS} = const., increase V_{DS} $\quad V_{DS} \in (V_t, V_{GS}) \quad (V_{DS} < V_{GS})$

→ As V_{DS} becomes larger, depth of the channel narrows due to E-field at the drain

↳ result: linear curves in the v-i characteristic bend



$$k_n = C_{ox} \mu_n \frac{W}{L}$$

$$k_n' = C_{ox} \mu_n$$

Let $V_{DS} \geq V_{OV}$ (saturation)

→ $V_{DS} = V_{OV}$: channel voltage drop is so large that the channel depth at the drain goes to zero

→ $V_{DS} > V_{OV}$: the channel depth won't increase anymore and there can't be more current flow even if V_{DS} keeps increasing

MOSFET regions of operation (N-channel, enhancement mode)

1) Triode / linear region : $v_{DS} < V_{ov}$ & $v_{GS} = V_t + V_{ov}$

$$i_D = k_n' \frac{W}{L} \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right] \quad \text{or}$$

$$i_D = k_n' \frac{W}{L} \left(V_{ov} - \frac{1}{2} v_{DS} \right) v_{DS}$$

$$g_{DS} = \frac{1}{r_{DS}} = k_n' \cdot \frac{W}{L} \cdot V_{ov}$$

2) Saturation region : $v_{DS} \geq V_{ov}$ & $v_{GS} = V_t + V_{ov}$

$$i_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_t)^2 \quad \text{or}$$

$$i_D = \frac{1}{2} k_n' \frac{W}{L} V_{ov}^2$$

$$\left[k_n = k_n' \frac{W}{L} \right]$$

3) $v_{GS} < V_t$: cutoff, no channel, no current flow

MOSFET regions of operation (P-channel, enhancement mode)

1) Triode / linear region : $v_{SD} < |V_{ov}|$ & $v_{SG} = |V_{tp}| + |V_{ov}|$

$$i_D = k_p' \frac{W}{L} \left[(v_{SG} - |V_{tp}|) v_{SD} - \frac{1}{2} v_{SD}^2 \right] \quad \text{or}$$

$$i_D = k_p' \frac{W}{L} \left(|V_{ov}| - \frac{1}{2} v_{SD} \right) v_{SD}$$

$$g_{DS} = \frac{1}{r_{DS}} = k_p' \frac{W}{L} |V_{ov}|$$

2) Saturation region : $v_{SD} \geq |V_{ov}|$

$$i_D = \frac{1}{2} k_p' \frac{W}{L} (v_{SG} - |V_{tp}|)^2 \quad \text{or}$$

$$i_D = \frac{1}{2} k_p' \frac{W}{L} V_{ov}^2$$

3) Cutoff : $v_{SG} < |V_{tp}|$: no current flow

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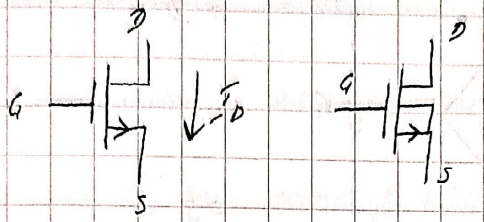
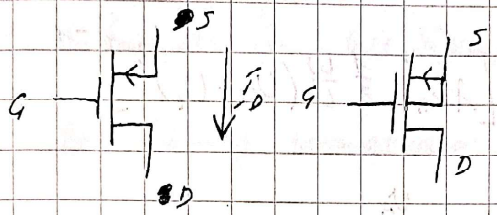
Note to those getting those notes. MOSFET symbols are "wrong". Different symbols are used in design or

Very bad circuit symbols, only good for seeing the direction of i_D

N-channel



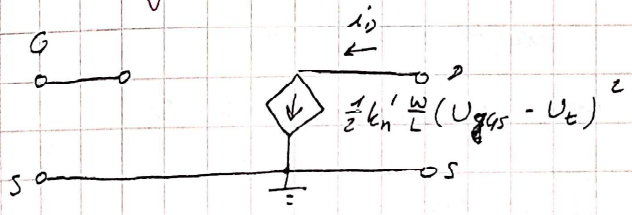
P-channel



For amplification, MOSFETS are operated in saturation;

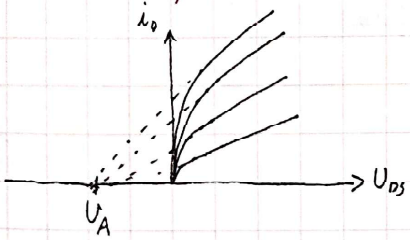
for switching: in the linear region

Small signal MOSFET model



The MOSFET in saturation is a voltage-controlled current source

Finite output resistance of the MOSFET



- previously assumed constant i_D in saturation
- a more accurate characteristic: i_D keeps

increasing a little bit in saturation region

- pinchoff region moves towards the source

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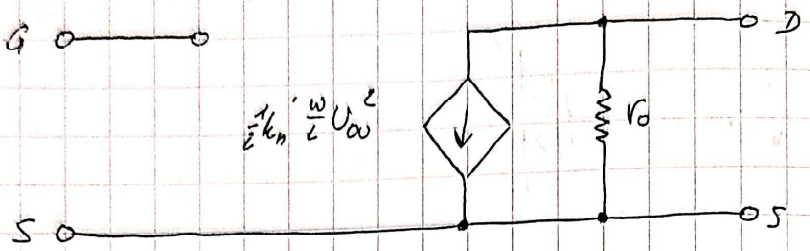
Improved MOSFET model:



$$r_o = \left[\frac{\partial i_d}{\partial v_{gs}}\right]_{v_{gs}=\text{const}} = \left[\lambda k_n' \cdot \frac{1}{2}\frac{W}{L}(V_{GS}-V_t)^2\right]^{-1}$$

v_{gs} = v_{GS}

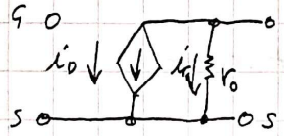
$$= \frac{1}{\lambda I_D} \Rightarrow r_o = \frac{V_A}{I_D} \qquad V_A: \text{early voltage}$$

Early voltage is given in the datasheet.

Updated drain current in saturation:

$$i_D = \frac{1}{2}k_n' \frac{W}{L}(V_{GS}-V_t)^2 \underbrace{(1+\lambda V_{DS})}_{\text{compensation factor}}$$

Ex. $V_{GS}=3V$, $V_t=0.5V$, $V_A=40V$, before $i_D = 31.25mA$ @ $V_{DS}=5V$



$$r_o = \frac{V_A}{i_D} = \frac{40V}{31.25mA} = 1.3k$$

i_D @5V $V_{DS} \Rightarrow i_{r_o} = \frac{5V}{1.3k} = 3.85mA$

$\therefore i_D = 31.25mA + 3.85mA = 35.1mA$

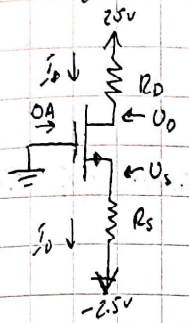
MOSFET at DC ... MOSFET biasing

→ apply high value DC inputs to the MOSFET

→ neglect channel modulation: $\lambda = 0$, $V_A \to \infty$

Ex. Select R_D and R_S such that: $I_D = 0.4mA$ and $V_D = 0.5V$.

$V_t = 0.7V$, $\mu_n C_{ox} = 100 \frac{\mu A}{V^2}$, $W = 32\mu m$, $L = 1\mu m$, $V_{DD} = 2.5V$, $V_{SS} = -2.5V$



Assuming saturation or triode, not in cutoff

$V_{OV} = V_{GS} - V_t = V_G - V_S - V_t = -V_S - 0.7V$

$V_{DS} = V_D - V_S = 0.5V - V_S$

Assuming saturation: $V_{DS} \geq V_{OV} \Rightarrow 0.5 - V_S \geq -V_S - 0.7V$

$\Rightarrow 0.5V > -0.7V$; yes, it's in saturation $\quad R = \frac{V}{I}$

$$R_D = \frac{2.5V - V_D}{I_D} = \frac{2.5V - 0.5V}{0.4mA} = \underline{5k}$$

R_S: $i_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 = 0.4mA \Rightarrow$ algebra

$$\Rightarrow V_{GS} = \sqrt{\frac{1}{50} V^2 \cdot \frac{40}{32}} + 0.7V = 1.2V$$

$V_{GS} = V_G - V_S \Rightarrow V_S = V_G - V_{GS} \Rightarrow V_S = 0 - 1.2V = -1.2V$

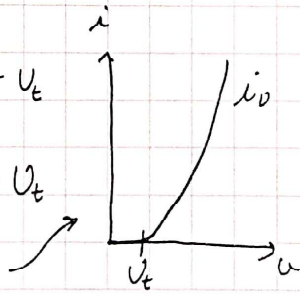
$$R_S = \frac{-1.2V - (-2.5V)}{0.4mA} = \underline{3.25k}$$

MOSFET as a diode

$v = V_{DS} = V_{GS}$, $V_{DS} \geq V_{OV} \Rightarrow V_D - V_S \geq V_G - V_S - V_t$

$\Rightarrow V_D \geq V_G - V_t$ or $V_D \geq V_G$ assuming $V_G > V_t$

$i_D = \frac{1}{2} k_n' \frac{W}{L} (v - V_t)^2$



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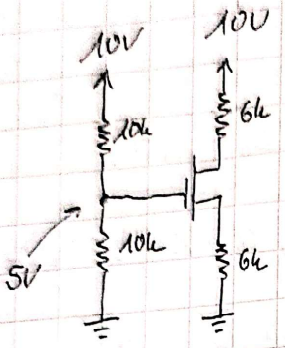
Example of MOSFET biasing (N-channel)

Find: I_D, V_{GS}, V_D, V_{DS}, V_S ; Given: $k_n' \frac{W}{L} = 1 \frac{mA}{V^2}$, $V_t = 1V$



$V_G = 5V$ from voltage divider

Assume saturation:

$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 = 0.5 (V_{GS} - 1V)^2 \frac{mA}{V^2}$

$V_S = I_D R_S = I_D \cdot 6k$

$\Rightarrow I_D = \frac{1}{2} \frac{mA}{V^2} (5V - I_D \cdot 6k - 1)^2 \Rightarrow 18 I_D^2 - 25 I_D + 8 = 0 \Rightarrow I_D = 0.5mA$ or $I_D = 0.89mA$

(mA and k cancel out)

if $I_D = 0.89mA \Rightarrow V_S = I_D R_S = 5.4V \Rightarrow V_{GS} = 5V - 5.4V < 0$ cutoff ✗

if $I_D = 0.5mA \Rightarrow V_S = 3V \Rightarrow V_{GS} = 5V - 3V = 2V > V_t$ not cutoff

↳ $V_{OV} = V_{GS} - V_t = 2V - 1V = 1V$

$V_D = V_{DD} - 6k I_D = 10V - 3V = 7V$

$V_{DS} = V_D - V_S = 7V - 3V = 4V$

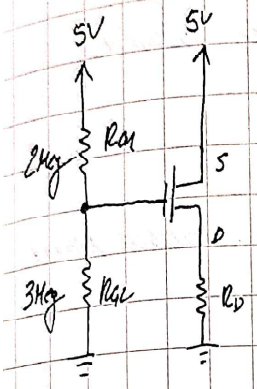
$V_{DS} > V_{OV}$ $(4V > 1V)$ ∴ saturation

Date

Example of MOSFET biasing (P-channel)

Choose: R_{G1}, R_{G2}, R_D ; gate voltages should be pretty large

Given: $I_D = 0.5mA$, $V_D = 3V$, $V_{tp} = -1V$, $k_p = 1 \frac{mA}{V^2}$, $\lambda = 0$



$V_S = 5V$, shorted to the supply rail

$I_D = \frac{1}{2} k_p (V_{SG} - |V_{tp}|)^2 \Rightarrow \sqrt{\frac{2I_D}{k_p}} = V_S - V_G - |V_{tp}|$

$\Rightarrow V_G = V_S - |V_{tp}| - \sqrt{\frac{2I_D}{k_p}} = 3V$

$V_{SG} = V_S - V_G = 5V - 3V = 2V$

$V_G = 5V \cdot \frac{R_{G2}}{R_{G1} + R_{G2}} = 3V \Rightarrow \frac{R_{G2}}{R_{G1} + R_{G2}} = \frac{3}{5}$

Let $R_{G2} = 3Meg \Rightarrow R_{G1} = 2Meg$

Check saturation: $V_{SG} > |V_{tp}| \Rightarrow 2V > 1V \Rightarrow$ not in cutoff

Saturation: $V_{SD} \geq |V_{ov}| \Rightarrow V_{SD} \geq |V_{SG} - |V_{tp}|| \Rightarrow 5V - 3V \geq |2 - |-1||$

$\Rightarrow 2V \geq 1V$ ... yes, it's in saturation

$R_D = \frac{3V}{0.5mA} = 6k\Omega$

Max R_D to keep the MOSFET in saturation: $V_{SD} = |V_{ov}| = 1V$

$V_S = 5V \Rightarrow V_D = 5V - 1V = 4V \Rightarrow R_{D\,max} = \frac{4V}{0.5mA} = 8k\Omega$

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Date

Bipolar junction transistor (BJT)

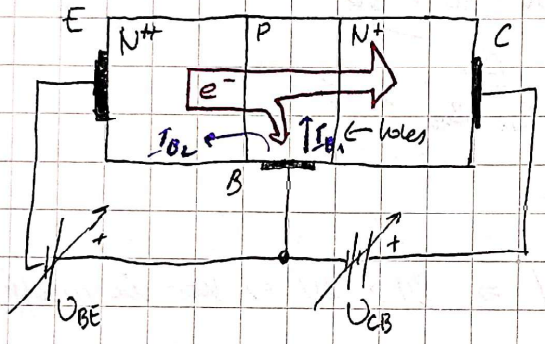
Bipolar: current flow composed of both electrons and holes

Junction: composed of 2 PN junctions

↳ operation depends on how junctions are biased

Current flows the most in active region.

Active region: BE junction forward biased, CB junction reversed biased



$i_E = i_B + i_C$

- majority carrier flow in both directions of the forward biased BE junction
- electrons in the base diffuse

Currents in active region

$$i_C = I_S e^{\frac{v_{BE}}{V_T}}$$

$$I_S = \frac{A_e q D_n n_i^2}{N_A W}$$

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Date

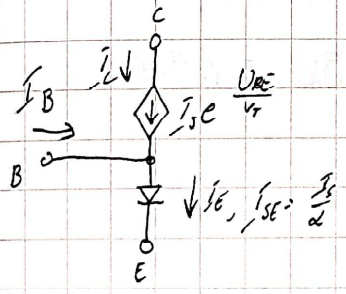
BJT in active region (NPN)

- $i_C = I_S e^{\frac{V_{BE}}{V_T}}$
- $i_B = \frac{i_C}{\beta}$
- $i_E = i_B + i_C = \frac{i_C}{\beta} + i_C = i_C\left(1+\frac{1}{\beta}\right) = \frac{\beta+1}{\beta} i_C$
- $i_C = \frac{\beta}{\beta+1} i_E = \alpha i_E$ $\quad \alpha = \frac{\beta}{\beta+1}$; $\beta = \frac{\alpha}{1-\alpha}$
- $i_C = \beta i_B = \alpha i_E$
- $i_B = (1-\alpha) i_E = \frac{i_E}{\beta+1}$

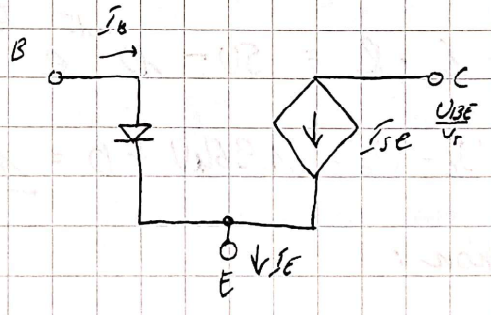
for PNP, replace V_{BE} with V_{EB}

Large signal NPN BJT models

Common base



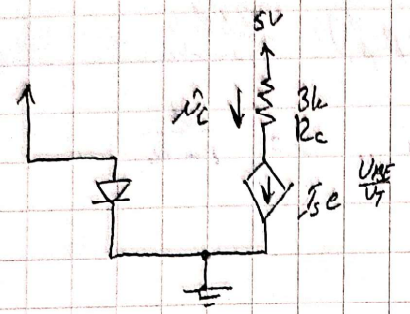
Common emitter



* Use this one when emitter is grounded

Example 6.1 : NPN BJT in active region

Given : $I_S = 10^{-15}\,A$, $\beta = 100$, $i_B = 10\mu A$, grounded emitter



Verify active region:

$$i_C = I_S e^{\frac{V_{BE}}{V_T}}$$

$$i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{\frac{V_{BE}}{V_T}} = 10\mu A$$

$$\Rightarrow e^{\frac{V_{BE}}{V_T}} = 10\mu A \cdot \frac{\beta}{I_S}$$

$$\Rightarrow \ln\left(e^{\frac{V_{BE}}{V_T}}\right) = \ln\left(10\mu A \cdot \frac{\beta}{I_S}\right)$$

$$\Rightarrow V_{BE} = V_T \ln\left(10\mu A \cdot \frac{\beta}{I_S}\right) \approx \underline{0.659\,V}$$

BE junction is forward biased ✓

$$V_C = 5V - i_C \cdot R_C = 5V - 10^{-15} e^{\frac{0.659}{26mV}} \cdot 3k = 1.946\,V$$

$$\Rightarrow V_{CE} = V_C - V_E = 1.964V - 0 = \underline{1.964\,V}$$

Active region:

- BE junction forward biased? $V_{BE} = 0.659\,V$ yes
- BC junction reverse biased? $V_{BC} = V_B - V_C = 0.659V - 2V < 0$ yes

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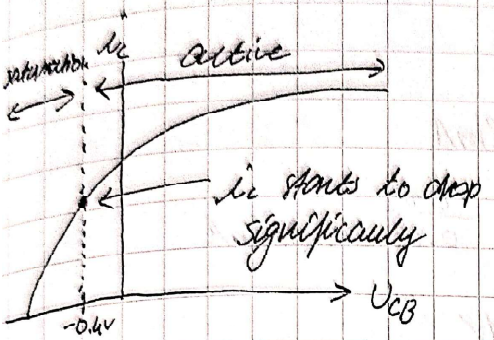
A more accurate way of defining forward bias

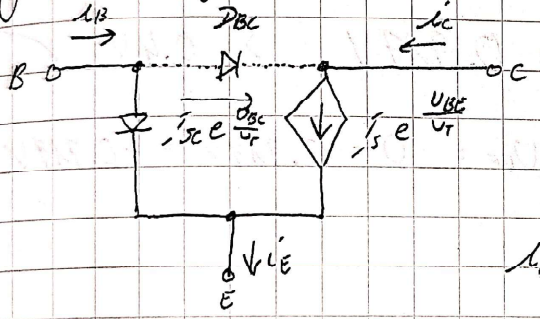
For active region, CB junction must be reverse biased

↳ more accurately: $V_{CB} \geq -0.4V$



V_{BE} is usually 0.7V

Why does i_C go to 0A in saturation or V_{CB} become more negative



$$I_S e^{\frac{V_{BC}}{V_T}} = i_C + I_{SC} e^{\frac{V_{BC}}{V_T}}$$

$$\Rightarrow i_C = I_S e^{\frac{V_{BE}}{V_T}} - I_{SC} e^{\frac{V_{BC}}{V_T}}$$

$$i_B = \frac{I_S}{\beta} e^{\frac{V_{BE}}{V_T}} + I_{SC} e^{\frac{V_{BC}}{V_T}}$$

β is no longer constant

β_{forced} implies saturation

$$\beta_{forced} = \frac{i_C}{i_B}\Big|_{saturation}, \quad \beta_{forced} \leq \beta$$

Actually determining NPN BJT operating region

- saturation: $V_{CE} < 0.3V$ (or $V_{CE} < V_{BE}$ [MIT])
- active: $V_{CE} > 0.3V$ & $V_{BE} > 0.4V$ (or $V_{BE} < V_{CE} < V_{CC}$ [MIT])

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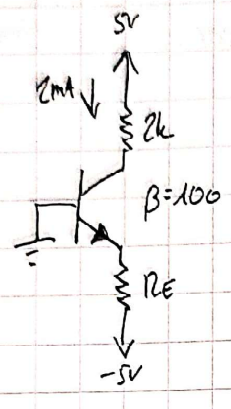
NPN BJT bias example



$V_{BE} = 0.7V$ @ $i_C = 1mA$, $i_B = 20\mu A$

Find: V_C, V_E, V_{CE}, i_C, i_E, R_E

Assume active region:

$i_C = \beta i_B = 100 \cdot 20\mu A = 2mA$

$i_E = \frac{i_C}{\alpha} = \frac{2mA}{\frac{\beta}{\beta+1}} = \frac{2mA}{0.99} = 2.02mA$

$V_C = V_{CC} - R_C i_C = 5V - 2k \cdot 2mA = 1V$

$I_S = i_C e^{-\frac{V_{BE}}{V_T}} = 1mA \cdot e^{-\frac{0.7}{25.9mV}} = 6.9 \cdot 10^{-16} A$

$V_{BE} = V_T \ln\left(\frac{2mA}{6.9 \cdot 10^{-16} A}\right) = 0.717V > 0.4V$ ✓

$V_{BE} = V_B - V_E \Rightarrow V_E = V_B - V_{BE} = 0 - 0.717V = -0.717V$

$R_E = \frac{V_E - V_{EE}}{i_E} = 2.12k$

Check active region: $V_{CE} > 0.3V$ & $V_{BE} > 0.4V$

$V_{BE} = 0.717V > 0.4V$ ✓

$V_{CE} = V_C - V_E = 1V - (-0.717V) = 1.717V > 0.3V$ ✓

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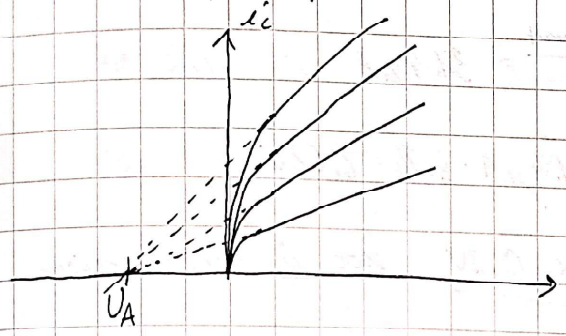
Collector current vs. collector-emitter voltage (i_C vs. V_{CE})

→ The curves are generated by varying V_{CE} and keeping V_{BE} constant

$$V_{CB} = V_{CE} - V_{BE} \qquad (V_C - \cancel{V_E} - V_B + \cancel{V_E})$$

→ As V_{CE} ↑, V_{CB} increases causing CB junction to be even further reverse biased

→ As V_{CE} ↓, V_{CB} approaches $-0.4V$ which will put it into saturation



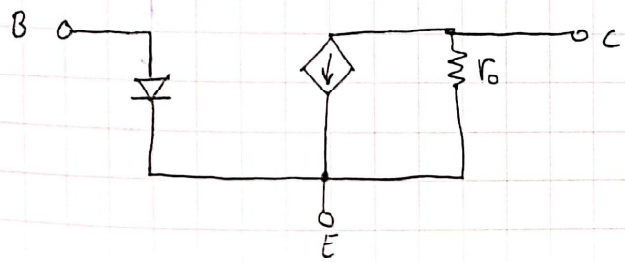
→ As V_{CE} ↑, i_C also slightly increases due to base width reduction from the expanding depletion region at CB junction (Early effect)

More accurately:

$i_C = I_S e^{\frac{V_{BE}}{V_T}} \left(1 + \frac{V_{CE}}{V_A}\right)$; The slope acts as output resistance r_o

$$r_o = \left[\frac{\partial i_C}{\partial V_{CE}}\Big|_{V_{BE}=\text{const.}}\right]^{-1} = \frac{\partial}{\partial V_{CE}}\left[I_S e^{\frac{V_{BE}}{V_T}}\left(1 + \frac{V_{CE}}{V_A}\right)\right] = I_S e^{\frac{V_{BE}}{V_T}} \cdot \frac{1}{V_A}$$

$$\therefore r_o = \frac{V_A}{I_C'}$$



BJT at DC

- cutoff : EBJ : RV CBJ : RV
- active : EBJ : FW CBJ : RV
- saturation : EBJ : FW CBJ : FW

FW: forward biased, RV: reverse biased

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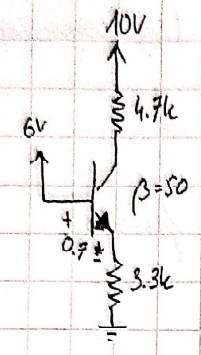
Example: NPN BJT in saturation



$V_{BE} = 0.7V$

$V_E = 6V - 0.7V = 5.3V$

$I_E = \frac{V_E}{3.3k} = \frac{5.3V}{3.3k} = 1.6mA$

$\alpha = \frac{\beta}{\beta+1} = \frac{50}{51} = 0.98$

$I_C = \alpha I_E = 0.98 \cdot 1.6mA = 1.57mA$

$I_C = \beta I_B \Rightarrow I_B = \frac{I_C}{\beta} = \frac{1.57mA}{50} = 31.4\mu A$

$V_C = 10 - I_C R_C = 10V - 31.4\mu A \cdot 4.7k = 2.62V$

$V_{CE} = 2.62V - 5.3V = -2.68V < 0.3V$ not in active region

$\Rightarrow$ Assume saturation

$V_{CE_{sat}} = 0.2V$; I_E and V_E are the same

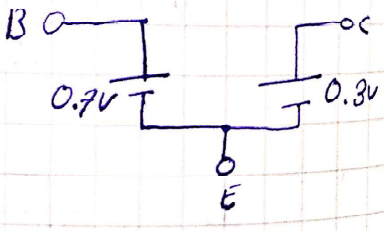
$V_C = V_E + V_{CE} = 5.3V + 0.2V = 5.5V$

$I_C = \frac{10 - V_C}{R_C} = 0.957mA$

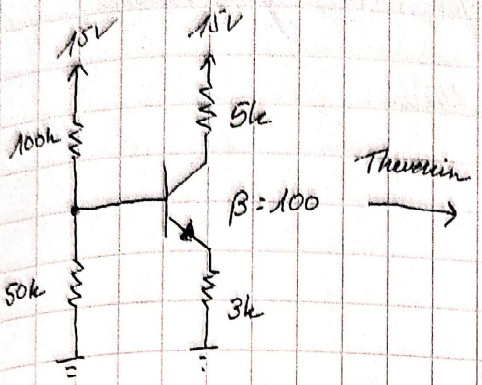
$I_B = I_E - I_C = 0.643mA$

$\Rightarrow \beta_{forced} = \frac{I_C}{I_B} = \frac{0.957}{0.643} = 1.5$

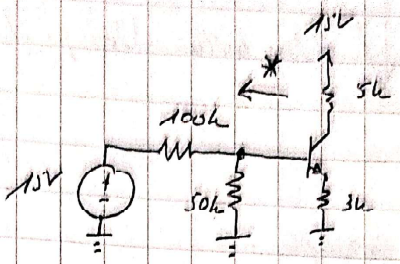
Model in saturation



Example 6.10 NPN BJT biasing (the worse one)



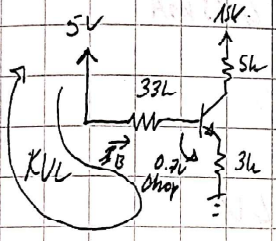
Thevenin →



*: Equivalent voltage and series resistance seen from the base

↳ Short the voltage source, then $R_{th} = 100k \parallel 50k = \underline{33k}$

$$U_{th} = 15V \cdot \frac{50k}{100k + 50k} = 15V \cdot \frac{1}{3} = \underline{5V}$$



(=)

KVL: $5V = I_B \cdot 33k + 0.7V + 3k \cdot I_E$

$$\Rightarrow 5V = \frac{I_E}{101} \cdot 33k + 0.7V + 3k \cdot I_E$$

$$\Rightarrow I_E = \frac{5V - 0.7V}{\frac{33k}{101} + 5k} = 1.29\,mA$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{1.29\,mA}{100 + 1} = 12.8\,\mu A$$

$$\Rightarrow I_E = \frac{U_E}{R_E} \Rightarrow U_E = I_E R_E = 3k \cdot 1.29mA = 3.87V$$

U_B: $U_B - U_E = 0.7V \Rightarrow U_B = U_E + 0.7V = 4.57V$

U_{CE}: $I_C = \frac{15V - U_C}{R_C} \Rightarrow U_C = 15V - I_C R_C \Rightarrow U_C = 15V - \beta I_B R_C = 8.6V$

$U_{CE} = U_C - U_E = 8.6V - 3.87V = 4.73V > 0.3V$

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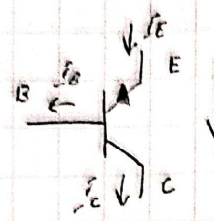
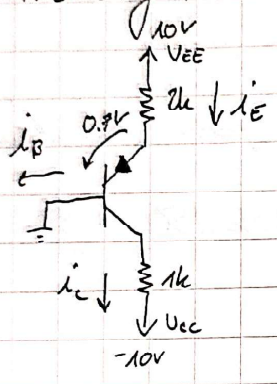
PNP BJT



Same equations as for NPN, just replace V_{BE} with V_{EB}, I think.

The only PNP example:



Assume active region

$V_{EB} = 0.7V \Rightarrow V_E - V_B = 0.7V \Rightarrow V_E - 0V = 0.7V$

$V_E = 0.7V$, $V_B = 0V$

$$i_E = \frac{V_{EE} - V_E}{R_E} = \frac{10V - 0.7V}{2k} = 4.65mA$$

$$i_C = \alpha i_E = \frac{100}{101} i_E = 4.6mA$$

$$V_C = V_{CC} + i_C R_C = -10V + 4.6mA \cdot 1k = -5.4V$$

Check active region:

$V_{EC} > 0.3V \Rightarrow V_E - V_C > 0.3V \Rightarrow 0.7V - (-5.4V) > 0.3V$

$\Rightarrow 6.1V > 0.3V$ yes, it's in active region

$$i_B = \frac{i_C}{\beta} = \frac{4.6mA}{100} = 46\mu A$$

A common application of a PNP BJT is in a push-pull amplifier